A

Project Report

on

PARIVA JEWELLERS

B-Tech-IT, Sem-VI

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CANDIDATE’S DECLARATION

We declare that the 6th semester project report entitled “PRIVA JEWELLERS” is our own work conducted under the supervision of Prof. Deepak. C. Vegda.

We further declare that to the best of our knowledge the report for B.Tech. VI semester does not contain part of the work which has been submitted either in this or any other university without proper citation.

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CERTIFICATE

This is to certify that the project carried out in the subject of Project-I, entitled "**PRIVA JEWELLERS**" and recorded in this report is a bonafide report of work of

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ABSTRACT

Jewellery shopping traditionally requires customers to visit physical stores to explore collections, compare prices, and make purchases. This process can be time-consuming and limits customers to the products available at a particular store. **Priva Jewellers** solves this problem by providing an online jewellery shopping platform that allows users to browse, select, and purchase jewellery products conveniently from anywhere.

The platform offers a secure and user-friendly shopping experience. Customers can explore products through category-based browsing and advanced filters such as price, metal type, purity, weight, and occasion. Each product page provides detailed information including price breakdowns such as base price, making charges, stone charges, and GST. The system also includes features such as user registration, OTP-based login verification, wishlist management, shopping cart functionality, coupon application, order tracking, invoice downloads, and secure payment options.

Unlike traditional systems, the platform integrates modern e-commerce features including secure online payments through Razorpay, personalized user profiles, and custom jewellery request functionality where customers can submit design requirements and receive price estimates from the administrator.

From a technical perspective, the system is developed using the MERN stack. The frontend is built using React.js with Vite, providing a fast and responsive user interface. The backend is developed using Node.js and Express.js to manage server-side operations and APIs, while MongoDB is used for storing user, product, and order data. Security features include JWT authentication and OTP-based email verification to ensure secure access and data protection.

Overall, **Priva Jewellers** enhances the jewellery shopping experience by combining traditional jewellery retail with modern web technologies. The platform provides customers with a convenient and secure way to explore and purchase jewellery online while enabling administrators to efficiently manage products, orders, users, and promotional offers.

1.INTRODUCTION

1.1 PROJECT DETAILS

i. Priva Jewellers is a web-based e-commerce platform developed to simplify the process of browsing, selecting, and purchasing jewellery products online.

ii. The website allows customers to explore a wide range of jewellery products categorized by type, metal, purity, weight, and occasion while providing detailed product information including price breakdown and product specifications.

iii. The platform provides secure authentication, wishlist functionality, cart management, order placement, and online payment using Razorpay. It also includes an administrative dashboard that allows administrators to manage products, orders, users, coupons, and custom jewellery requests efficiently.

1.2 PURPOSE

i. The primary purpose of the **Priva Jewellers website** is to provide customers with a convenient and secure platform for purchasing jewellery products online.

ii. The system aims to enhance customer experience by providing advanced search filters, detailed product information, secure payment options, and order tracking features while enabling administrators to efficiently manage the store operations.

1.3 SCOPE

i. The scope of the Priva Jewellers system includes the development of a full-stack e-commerce website that supports user registration, authentication, product browsing, wishlist management, shopping cart functionality, order placement, and secure online payments.

ii. The platform also includes an admin management system that allows administrators to manage products, banners, coupons, orders, reviews, users, and custom jewellery design requests. The website is designed to be responsive and accessible across different devices.

1.4 OBJECTIVE

i. The main objective of the **Priva Jewellers project** is to develop a secure, user-friendly, and efficient online jewellery shopping platform that enhances customer engagement and simplifies the buying process.

ii. The system also aims to provide powerful administrative tools for managing inventory, monitoring orders, analyzing sales data, and handling custom jewellery requests, thereby improving overall business efficiency.

1.5 TECHNOLOGY AND LITERATURE REVIEW

1.5.1 Technology

i. The **Priva Jewellers website** is developed using the MERN Stack, which includes MongoDB, Express.js, React.js, and Node.js, enabling efficient full-stack web application development.
ii. The frontend is built using React with Vite, along with supporting libraries such as React Router for navigation, Axios for API communication, Recharts for analytics visualization, Swiper for interactive product displays, and Leaflet for map-based address selection.
iii. The backend is developed using Node.js and Express.js, which handle server-side logic, authentication, API endpoints, and communication with the MongoDB database through Mongoose.
iv. Secure online payments are integrated using Razorpay, enabling customers to complete transactions through UPI, debit cards, credit cards, and net banking.
v. Additional backend technologies include JWT for authentication and role-based access control, Nodemailer for email notifications and OTP verification, Multer for file uploads, and Puppeteer for generating downloadable PDF invoices.

1.5.2 Literature Review

i. The development of the **Priva Jewellers platform** was influenced by research and studies related to modern e-commerce systems, secure payment integration, and full-stack web application development using the MERN stack.
ii. The design of the system follows best practices in user interface design, secure authentication mechanisms, and scalable backend architecture, ensuring an efficient and user-friendly experience.
iii. The use of React for frontend development and Node.js with Express for backend services provides a scalable and maintainable architecture, as widely recommended in modern web development frameworks.

- **MERN Stack:** Efficient full-stack development with JavaScript technologies.

- **Razorpay:** Secure and reliable online payment gateway integration.
- **E-commerce Systems:** Modern trends in digital retail platforms and online transaction systems.

2.PROJECT MANAGEMENT

2.1 FEASIBILITY STUDY

2.1.1 Technical Feasibility

i. **Technology:** The Priva Jewellers website is developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), which is widely used for modern full-stack web applications.
ii. **Scalability:** MongoDB provides a flexible and scalable database solution, allowing the system to efficiently handle large numbers of users, products, and transactions.
iii. **Security:** Security is ensured using JWT-based authentication, OTP verification via email, and secure payment integration using Razorpay, which protects user data and financial transactions.

2.1.2 Time Schedule Feasibility

i. **Development Time:** The estimated time required for the development of the Priva Jewellers website is approximately 4 months, including design, development, testing, and deployment.
ii. The timeline allows sufficient time for implementing user features, admin functionalities, payment integration, and testing.

2.1.3 Operational Feasibility

i. User Adoption**:** The system is designed to be simple and user-friendly, enabling customers to easily browse jewellery products, place orders, and track purchases.
ii. The platform is expected to improve the online shopping experience and attract more customers through digital accessibility and convenience.

2.1.4 Implementation Feasibility

i. Development Team Members**:** The project is implemented by team members skilled in MERN stack technologies, including React for frontend development and Node.js with Express for backend development.
ii. The team possesses the required knowledge of MongoDB database management, payment gateway integration, and web application development.

2.2 PROJECT PLANNING

2.2.1 Project Development Approach and Justification

i. **Agile Development:** The project follows an Agile development methodology that allows continuous improvements and flexibility during development.
ii. **Justification:** Agile methodology helps in adapting new features, improving system performance, and incorporating feedback during development stages.

2.2.2 Project Plan

i. **Phase 1:** Planning and designing the user interface and website layout.
ii. **Phase 2:** Development of core features such as user authentication, product browsing, cart management, and admin dashboard.
iii. **Phase 3:** Integration of payment gateway, testing of functionalities, and final deployment.

2.2.3 Milestones and Deliverables

- **Milestone 1:** Project planning and requirement analysis completed.
- **Milestone 2:** Design and frontend interface development completed.
- **Milestone 3:** Backend APIs and database integration completed.
- **Milestone 4:** Payment gateway integration and testing completed.
- **Milestone 5:** Final system testing and deployment.

2.2.4 Roles and Responsibilities

i. **Frontend Developer** – Responsible for designing the user interface, implementing product pages, filters, cart, and responsive website layout using React.
ii. **Backend Developer** – Responsible for developing APIs, database integration, authentication system, payment integration, and server-side logic using Node.js, Express, and MongoDB.

2.2.5 Group Dependencies

i. **Technical Dependencies:** React.js, Node.js, Express.js, MongoDB, Razorpay payment gateway, Nodemailer, JWT authentication, and other supporting libraries used in the MERN stack.

2.3 PROJECT SCHEDULING

Table 2.1 Project Scheduling

Milestone	**Start Date**	**End Date**
Authentication System	06/12/2025	05/01/2026
Product & Database Integration	07/01/2026	15/02/2026
Cart, wishlist & Orders	18/02/2026	25/02/2026
Razorpay Payment Integration	27/02/2026	03/03/2026
Admin Dashboard & Testing	05/03/2026	09/03/2026

3 SYSTEM REQUIREMENTS STUDY

3.1 STUDY OF CURRENT SYSTEM

i. **Current System:** Traditional jewellery shopping is mostly done through physical jewellery stores where customers must visit the store to view and purchase products.

ii. **Functionality:** Customers browse jewellery items displayed in the store, inquire about product details from store staff, and make purchases directly at the store location.

iii. Some jewellery businesses also use basic websites or social media platforms to display products, but these systems often lack full e-commerce functionality such as secure payments, order tracking, and automated management.

3.2 PROBLEM AND WEAKNESS OF CURRENT SYSTEM

i. **Limited Accessibility:** Customers must visit the physical store to view jewellery products, which can be time-consuming and inconvenient.

ii. **Limited Product Information:** Traditional systems often provide limited product details and lack advanced search or filtering options.

iii. **Manual Management:** Inventory, orders, and customer records are often managed manually, which increases the chances of errors.

iv. **Lack of Online Transactions:** Many traditional systems do not support secure online payments or digital order processing.

3.3 USER CHARACTERISTICS

i. **Customers:** Individuals interested in purchasing jewellery products online. They require a simple interface to browse products, add items to the cart, place orders, and track their purchases.

ii. **Administrators:** Store managers or administrators responsible for managing products, orders, users, coupons, and other system functionalities.

3.4 HARDWARE AND SOFTWARE REQUIREMENTS

i. **Minimum Hardware:**
A computer, laptop, tablet, or smartphone with internet connectivity capable of accessing modern web applications.

ii. **Software Requirements:**
Modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari.

iii. **Server Requirements:**
Node.js runtime environment for backend processing and MongoDB database for storing application data.

iv. **Payment GatewayIntegration:**
The system supports integration with Razorpay payment gateway to allow secure online payments through UPI, debit cards, credit cards, and net banking.

3.5 CONSTRAINTS

3.5.1 Regulatory Policies

The system must comply with data protection and online transaction regulations to ensure safe handling of customer information and payment details.

3.5.2 Hardware Limitations

System performance may depend on the user’s device specifications and internet connectivity.

3.5.3 Interfaces to Other Applications

The system integrates with third-party services such as Razorpay for payment processing and email services for OTP verification and notifications.

3.5.4 Parallel Operations

The system should support multiple users simultaneously performing operations such as browsing products, placing orders, and managing accounts.

3.5.5 Higher Order Language Requirements

The system is developed using JavaScript technologies including React.js for frontend and Node.js with Express.js for backend development, along with MongoDB for database management.

3.5.6 Reliability Requirements

The system must provide reliable service with minimal downtime and ensure the integrity of stored data.

3.5.7 Criticality of the Application

The application is important for supporting online jewellery sales, secure transactions, and efficient order management.

3.5.8 Safety and Security Consideration

Security measures include JWT authentication, OTP-based login verification, secure payment processing, and database protection mechanisms.

3.6 Assumption and dependencies

i. **Assumptions:**
   Users accessing the Priva Jewellers website have internet connectivity and devices capable of running modern web browsers.
ii. The system assumes that users are familiar with basic online shopping operations such as browsing products, adding items to the cart, and completing payments online.
iii. The system also depends on third-party services such as Razorpay for payment processing and email services for OTP verification and notifications.

4. SYSTEM ANALYSIS

4.1 REQUIREMENTS OF NEW SYSTEM (SRS)

4.1.1 User Requirements

A. General User (Customer) Requirements

- Users must be able to register an account using email and password.
- Users must be able to log in securely with OTP verification.
- Users should be able to browse jewellery products through categories and search options.
- Users should be able to filter products by price, metal type, purity, weight, and occasion.
- Users must be able to view detailed product information including price breakdown.
- Users should be able to add products to cart and wishlist.
- Users should be able to place orders using Cash on Delivery or online payment via Razorpay.
- Users must be able to track order status and view order history.
- Users should be able to download invoices for placed orders.
- Users should be able to submit reviews and ratings for delivered products.
- Users must be able to submit custom jewellery design requests with images.
- Users should be able to manage their profile and delivery addresses.

B. Administrator Requirements

- Admin must be able to log in securely with password and email OTP verification.
- Admin should have access to an admin dashboard displaying system statistics and analytics.
- Admin must be able to add, update, and delete jewellery products.
- Admin should be able to manage product images and categories.
- Admin must be able to manage orders and update order status.
- Admin should be able to create and manage discount coupons.
- Admin must be able to view and manage registered users and block/unblock accounts if required.
- Admin should be able to manage banners and promotional content.
- Admin must be able to review and respond to custom jewellery **design** requests.
- Admin should be able to monitor system activities through dashboard analytics.

4.1.2 System Requirements

4.1.2.1Functional Requirements

FR1: User Authentication

1.1 User Registration

Description: New users can create an account on the platform.
• **IP:** Name, email, password, mobile number.
• **OP:** User account successfully created.
• **Processing:** System validates input data, sends OTP to email for verification, and stores user details in the database.

1.2 User Login

Description: Registered users can log in to the system.
• **IP:** Email and password.
• **OP:** User successfully logged into the system.
• **Processing:** System verifies credentials and authenticates the user using JWT.

1.3 Forgot Password

Description: Users can reset their password if forgotten.
• **IP:** Registered email address.
• **OP:** Password reset link or OTP sent to email.
• **Processing:** System verifies email and allows the user to create a new password.

1.4 Profile Management

Description: Users can view and update their profile information.
• **IP:** Updated profile details (name, phone number, etc.).
• **OP:** Updated profile information displayed.
• **Processing:** System validates and updates user data in the database.

FR2: Product Browsing

2.1 View Product Catalog

Description: Users can browse jewellery products from the catalog.
• **IP:** Category selection or homepage browsing.
• **OP:** List of products displayed.
• **Processing:** System retrieves product data from the database.

2.2 Product Search

Description: Users can search for products.
• **IP:** Search keyword entered by user.
• **OP:** Matching products displayed.
• **Processing:** System queries database using search filters.

2.3 Product Filtering and Sorting

Description: Users can filter and sort products.
• **IP:** Filter values (price, metal, occasion, weight).
• **OP:** Filtered product list displayed.
• **Processing:** System applies filtering logic and displays results.

FR3: Cart Management

3.1 Add Product to Cart

Description: Users can add products to their shopping cart.
• **IP:** Selected product and quantity.
• **OP:** Product added to cart successfully.
• **Processing:** System stores cart item in database.

3.2 Update Cart

Description: Users can update product quantity in the cart.
• **IP:** Updated quantity value.
• **OP:** Cart updated successfully.
• **Processing:** System updates cart data in database.

3.3 Remove Product from Cart

Description: Users can remove items from the cart.
• **IP:** Product selected for removal.
• **OP:** Product removed from cart.
• **Processing:** System deletes item from cart collection.

FR4: Wishlist Management

4.1 Add Product to Wishlist

Description: Users can save products in their wishlist.
• **IP:** Selected product.
• **OP:** Product added to wishlist.
• **Processing:** System stores wishlist item in database.

4.2 Remove Product from Wishlist

Description: Users can remove products from wishlist.
• **IP:** Product selected for removal.
• **OP:** Product removed from wishlist.
• **Processing:** System deletes item from wishlist.

FR5: Order Management

5.1 Checkout Process

Description: Users can place an order for products in the cart.
• **IP:** Cart items and delivery address.
• **OP:** Order successfully created.
• **Processing:** System creates order record and updates inventory.

5.2 Order Tracking

Description: Users can track the status of their orders.
• **IP:** Order ID.
• **OP:** Order status displayed (pending, shipped, delivered).
• **Processing:** System retrieves order details from database.

5.3 Order Cancellation

Description: Users can cancel orders before shipment.
• **IP:** Order cancellation request.
• **OP:** Order marked as cancelled.
• **Processing:** System updates order status.

FR6: Payment Processing

6.1 Online Payment

Description: Users can make payments using Razorpay.
• **IP:** Payment method (UPI, debit card, credit card).
• **OP:** Payment confirmation displayed.
• **Processing:** System sends payment request to Razorpay and verifies transaction.

6.2 Cash on Delivery

Description: Users can choose COD for order payment.
• **IP:** COD option selected during checkout.
• **OP:** Order placed successfully.
• **Processing:** System records payment mode as COD.

FR7: Product Reviews

7.1 Submit Review

Description: Users can submit reviews for delivered products.
• **IP:** Rating and review comment.
• **OP:** Review displayed on product page.
• **Processing:** System validates order status and stores review.

FR8: Custom Jewellery Request

8.1 Submit Custom Design Request

Description: Users can request custom jewellery designs.
• **IP:** Design description, image upload, budget details.
• **OP:** Custom request submitted successfully.
• **Processing:** System stores request and notifies admin.

FR9: Admin Management

9.1 Product Management

Description: Admin can add, update, and delete products.
• **IP:** Product details (name, price, category, images).
• **OP:** Product updated in system.
• **Processing:** System updates product collection in database.

9.2 Order Management

Description: Admin can view and update order status.
• **IP:** Order status update.
• **OP:** Updated order status visible to users.
• **Processing:** System updates order record.

9.3 User Management

Description: Admin can view and manage user accounts.
• **IP:** User management actions (block/unblock).
• **OP:** User account status updated.
• **Processing:** System updates user status in database.

4.1.2.2Non-Functional Requirements

- **Performance**
  The system should provide fast response time while browsing products, searching items, and placing orders. It should handle multiple users simultaneously without significant delays.

- **Security**
  The system must protect user data through secure authentication mechanisms such as JWT authentication and OTP verification. Passwords should be encrypted, and payments must be processed securely using the Razorpay payment gateway.

- **Reliability**
  The system should operate reliably and ensure accurate storage and retrieval of user data, product information, and order details.

- **Usability**
  The website should provide a simple and user-friendly interface so that customers can easily navigate, browse products, and complete purchases.

- **Scalability**
  The system should be capable of handling an increasing number of users, products, and transactions as the business grows.

- **Compatibility**
  The application should work correctly on different web browsers such as Chrome, Firefox, Edge, and Safari, and across different devices.

- **Maintainability**
  The system should be designed in a modular way so that future updates, bug fixes, and improvements can be implemented easily.

- **Availability**
  The system should remain accessible to users with minimal downtime to ensure uninterrupted service.

4.2 FEATURES OF THE NEW SYSTEM

The new system, **Priva Jewellers**, introduces several important features to improve the online jewellery shopping experience for customers and administrators:

- **Secure Authentication System:** User registration and login with email OTP verification and secure password protection.
- **Online Jewellery Marketplace:** Users can browse and purchase jewellery products through an organized catalog and category-based navigation.
- **Advanced Product Search:** Users can search products with filters such as price, metal type, purity, weight, and occasion.
- **Wishlist & Cart System:** Users can save products in wishlists and manage items in their shopping cart before placing orders.
- **Secure Payment Integration:** Supports online payments through Razorpay as well as Cash on Delivery (COD).
- **Order Management & Tracking:** Users can view order history, track orders, and download invoices.
- **Custom Jewellery Requests:** Users can submit custom jewellery design requests with images and descriptions.
- **Product Reviews & Ratings:** Customers can provide ratings and reviews for purchased products.
- **Admin Dashboard:** Admins can manage products, users, orders, coupons, banners, and custom design requests.
- **Email Notifications:** Automated emails are sent for OTP verification, order confirmation, and request updates.

4.3 NAVIGATION CHART

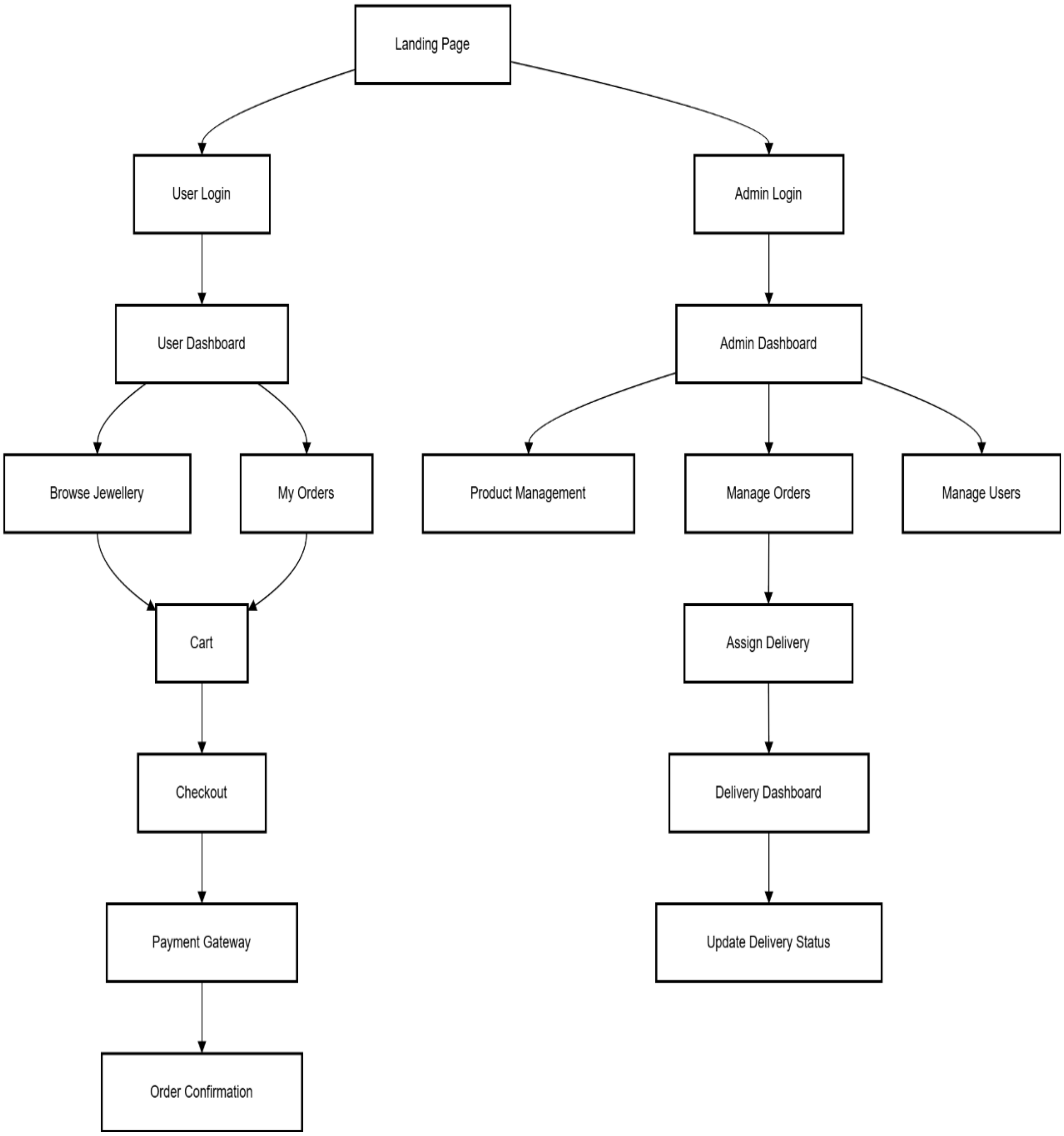


Fig.4.1 Navigation chart

4.4 CLASS DIAGRAM

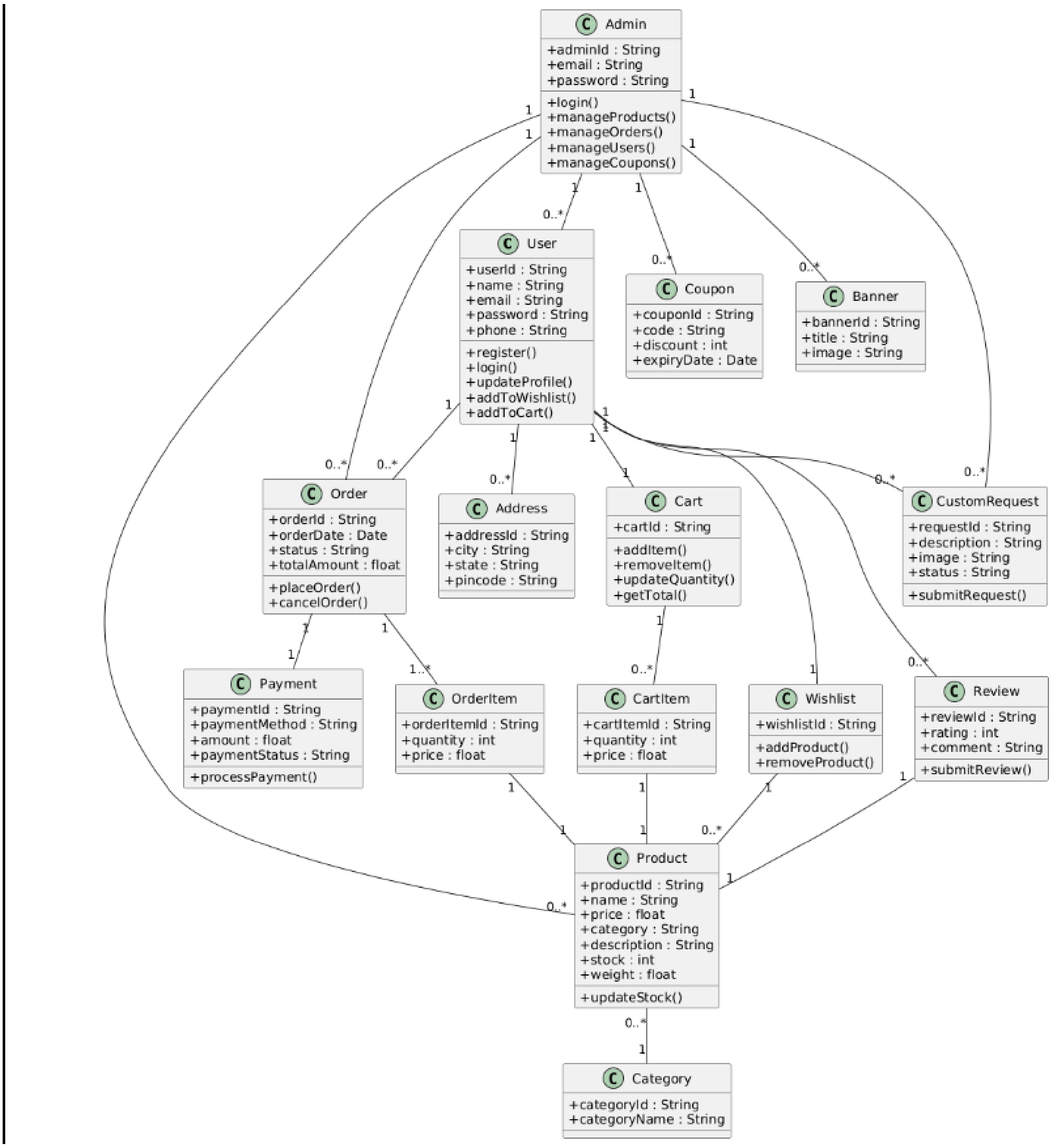


Fig 4.2 Class Diagram

4.5 USE CASE DIAGRAM

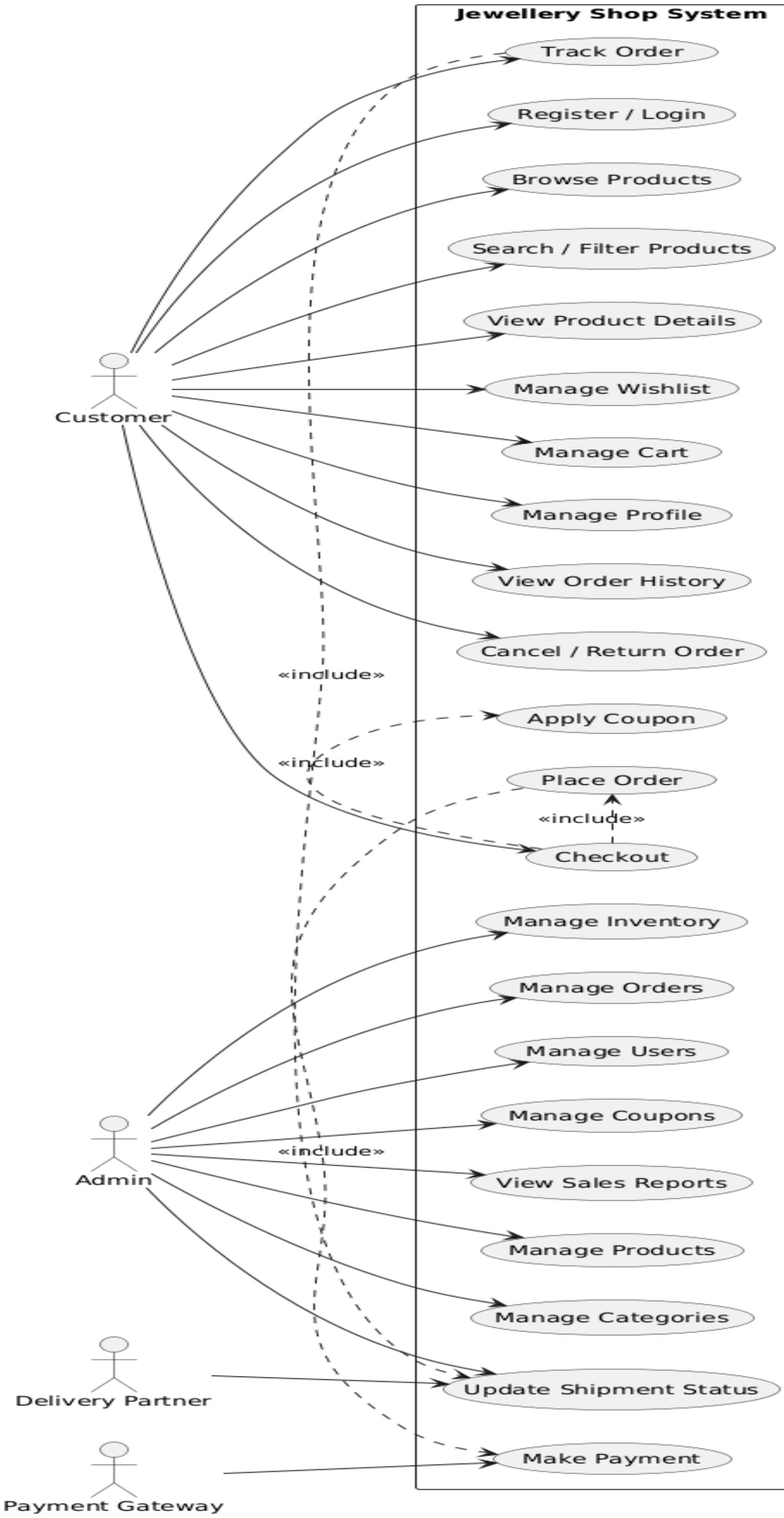


Fig 4.3 Use case diagram

4.6 SEQUENCE DIAGRAM

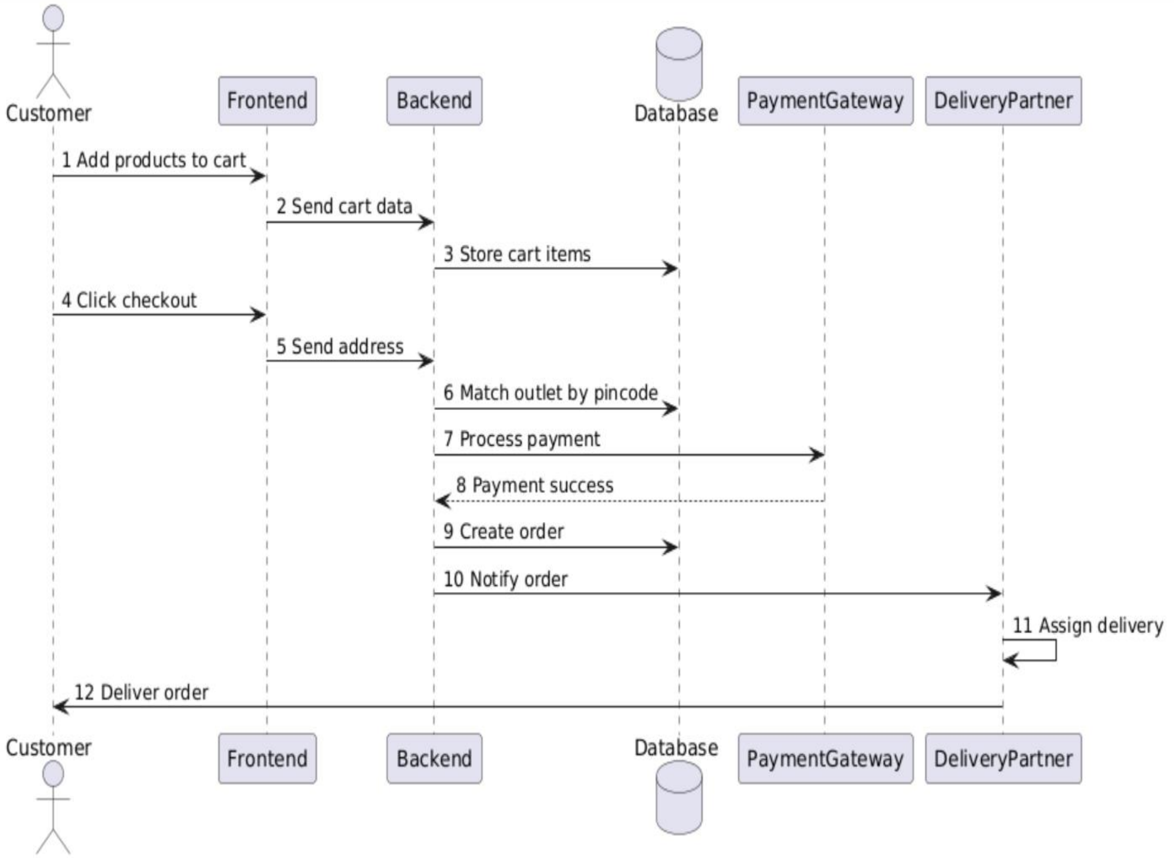


Fig 4.4 Sequence Diagram

4.7 ER DIAGRAM

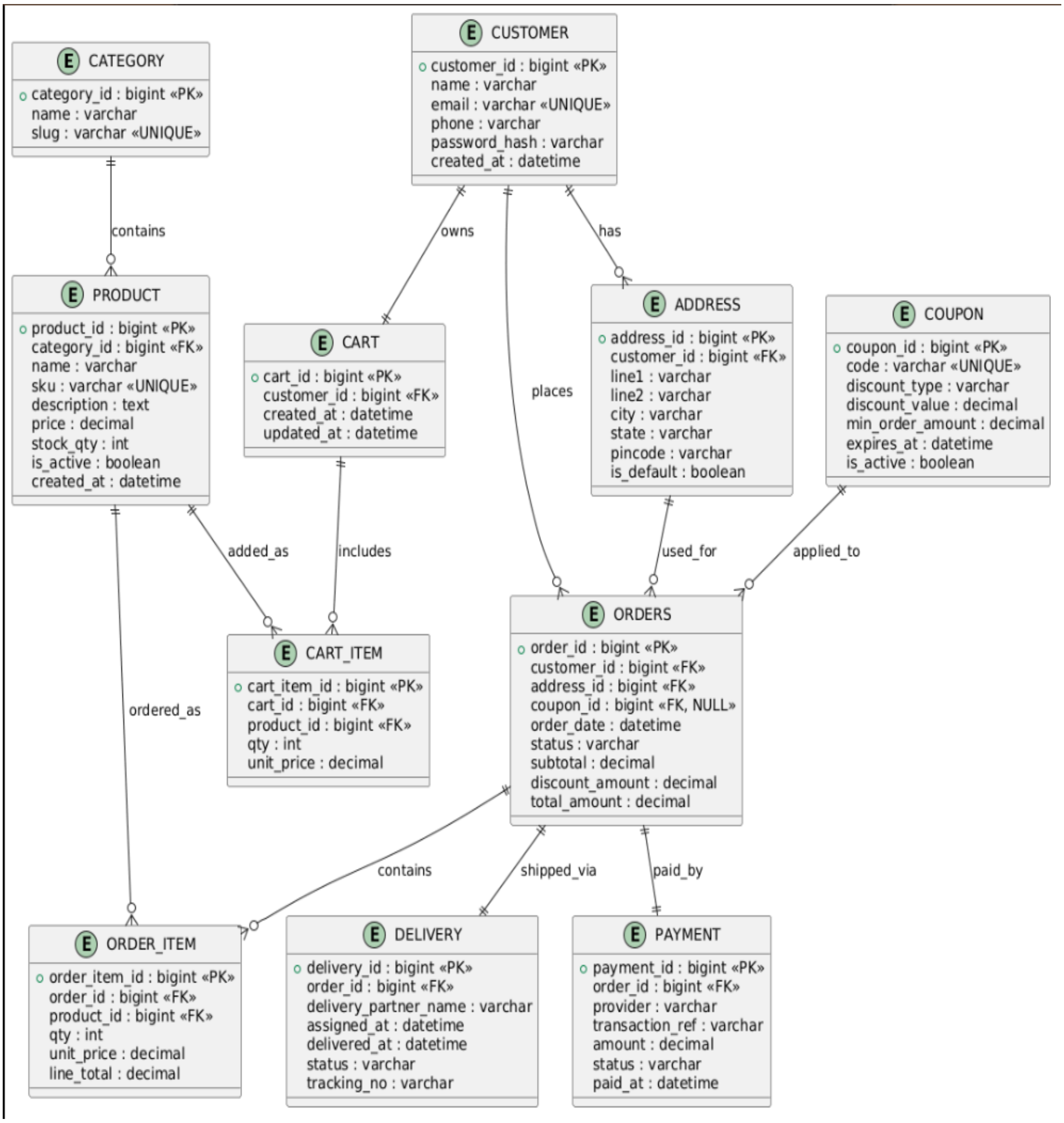


Fig 4.5 ER Diagram

5.SYSTEM DESIGN

5.1 System Architecture Design

The **Priva Jewellers Management System** follows a **three-tier architecture** that separates the application into presentation, application, and data layers. This architecture helps in improving scalability, maintainability, and performance of the system.

Three-Tier Architecture Layers

1. Presentation Layer

- Implemented using React.js
- Responsible for user interface and interaction
- Allows users to browse jewellery products, manage inventory, and generate bills.

2. Application Layer

- Implemented using Express.js & Node.js
- Handles business logic such as product management, billing, authentication, and order processing.
- Processes requests from the frontend and communicates with the database.

3. Data Layer

- Implemented using Mongodb
- Stores information related to jewellery products, customers, transactions, and inventory.

Advantages of this Architecture

- Separation of concerns
- Better scalability
- Easier maintenance and debugging
- Improved security and performance

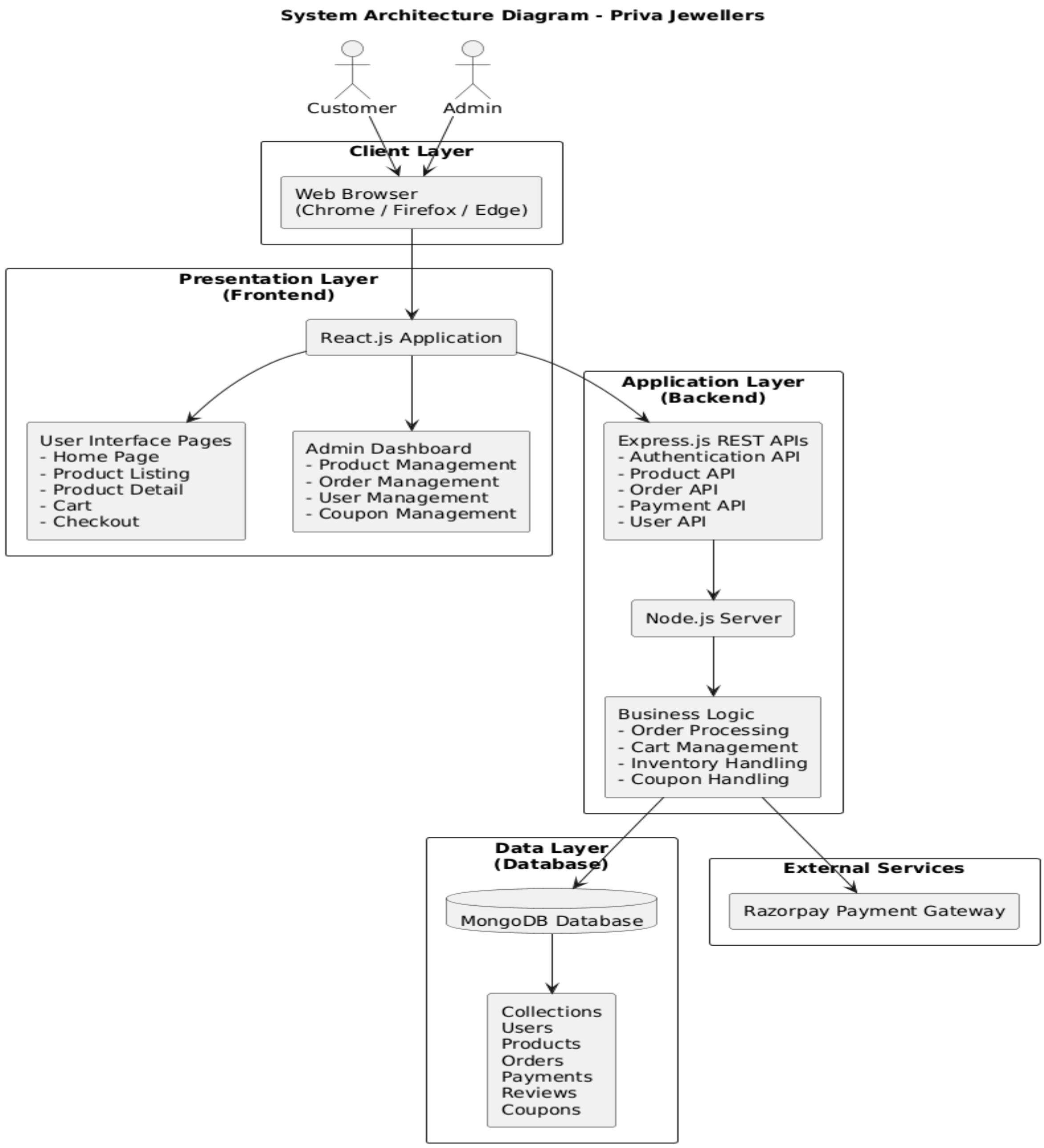


Fig 5.1 System Architecture Diagram

- The Priva Jewellers Management System follows a three-tier architecture consisting of the presentation layer, application layer, and data layer. The presentation layer provides the user interface using React. The application layer handles business logic such as authentication, inventory management, and billing using Express APIs. The data layer stores all persistent information including products, customers, and transactions in a MongoDB database.

5.1.1 COMPONENT DIAGRAM

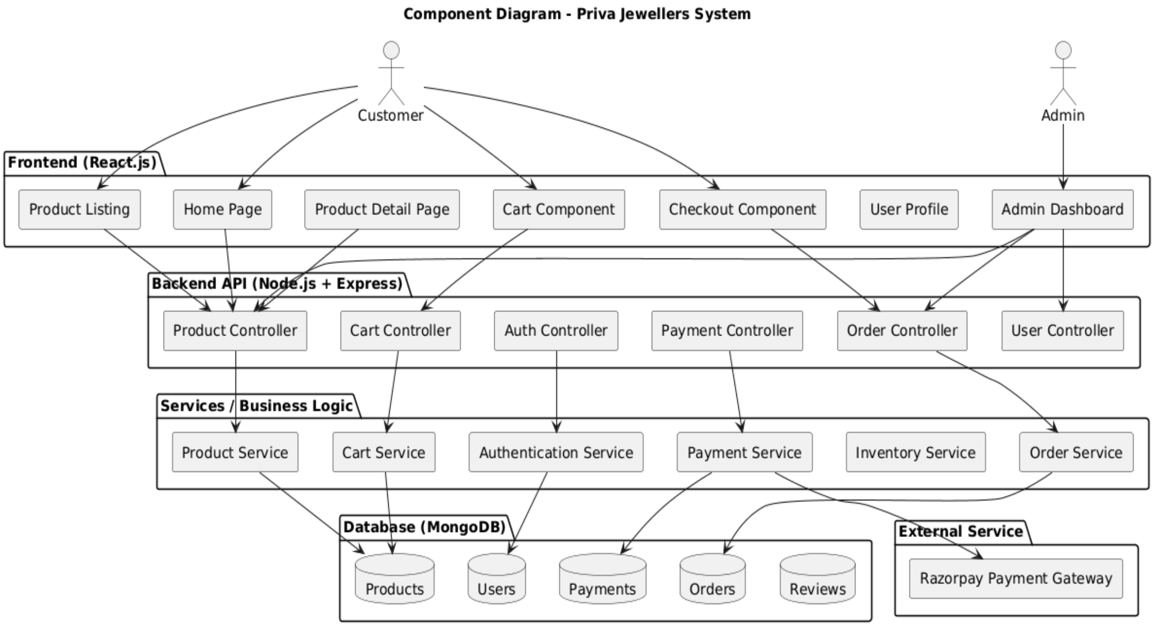


Fig 5.2 Component Diagram

- The component diagram of the Priva Jewellers Management System illustrates the interaction between the main system components. The user interface built using React communicates with backend services implemented using Express.js & Node.js. The backend includes modules such as authentication, product management, customer management, billing, and inventory management. These components interact with the MongoDB database to store and retrieve system data.

5.1.2 DEPLOYMENT DIAGRAM

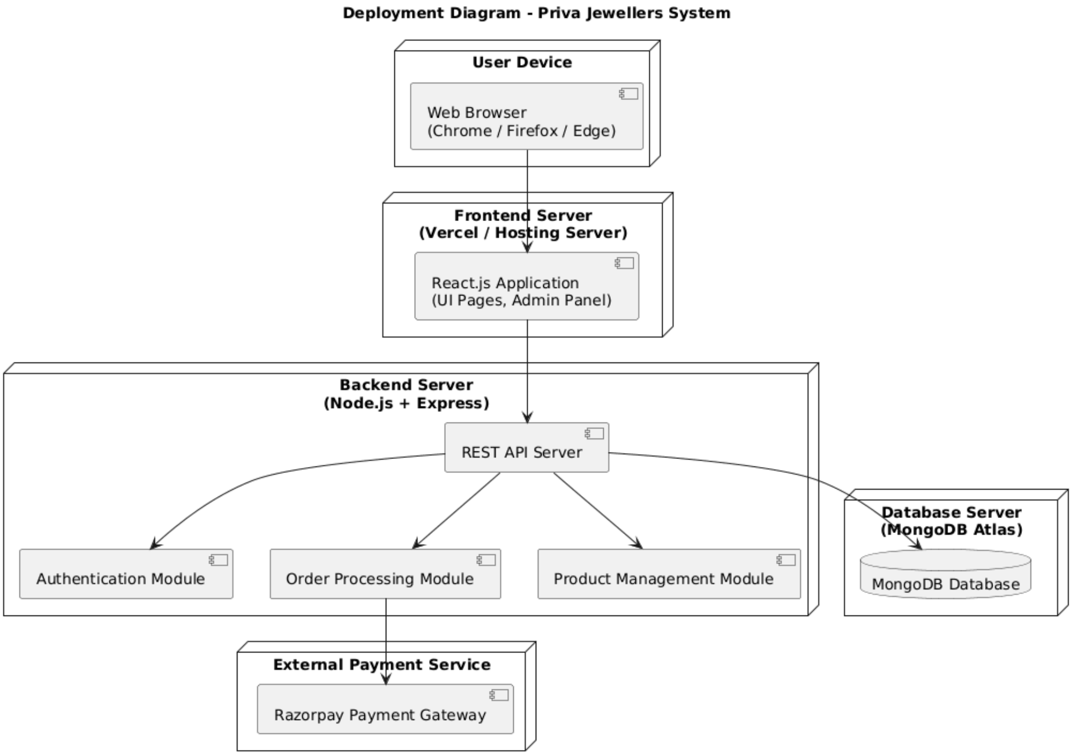


Fig 5.3 Deployment Diagram

- The deployment diagram represents the physical architecture of the Priva Jewellers Management System. Users access the system through a web browser on a client device. The frontend application built with React communicates with the backend server developed using Express.js & Node.js. The backend processes requests and interacts with the MongoDB database for storing and retrieving system data. The system can also connect to external payment gateways to process transactions securely.

Table 5.1 Deployment Environment

Layer	Technology
Client	Web Browser (Chrome, Firefox, Edge)
Frontend	React.js + Vite
Backend	Node.js + Express.js
Database	MongoDB
Authentication	JWT + Email OTP Verification
Payment Gateway	Razorpay
API Communication	REST APIs using Axios
File Upload	Multer
Email Service	Nodemailer
Hosting Platform	Vercel / Render / Railway

5.2 DATABASE DESIGN

5.2.1 Table and Relationship

Table 5.2 main database collection

Table Name	Description
Users	Stores customer, admin, and outlet manager information
Products	Stores jewellery product details
Orders	Stores customer order information
OrderItems	Stores products included in each order
Payments	Stores payment transaction details
DeliveryPartners	Stores delivery personnel information

Table 5.3 Users Table

Field Name	Data Type	Description
userId	ObjectId	Unique user ID
name	String	Name of user
email	String	User email
password	String	Hashed password
role	String	Customer / Admin / OutletManager
phone	String	Contact number
address	String	User address

Table 5.4 Products Table

Field Name	Data Type	Description
productId	ObjectId	Unique product ID
name	String	Jewellery product name
category	String	Product category
price	Number	Product price
stockQuantity	Number	Available quantity
imageURL	String	Product image
description	String	Product details

Table 5.5 Orders Table

Field Name	Data Type	Description
orderId	ObjectId	Unique order ID
customerId	ObjectId	Reference to Users
outletId	ObjectId	Reference to Outlets
orderDate	Date	Date of order
totalAmount	Number	Total order price
orderStatus	String	Current order state

Table 5.6 OrderItems Table

Field Name	Data Type	Description
orderItemId	ObjectId	Unique order item ID
orderId	ObjectId	Reference to Orders
productId	ObjectId	Reference to Products
quantity	Number	Quantity ordered
price	Number	Product price

Table 5.7 Payments Table

Field Name	Data Type	Description
paymentId	ObjectId	Unique payment ID
orderId	ObjectId	Reference to Orders
paymentMethod	String	Card / UPI / COD
paymentStatus	String	Paid / Pending / Failed
transactionId	String	Payment gateway transaction ID

Table 5.8 Delivery Boys Table

Field Name	Data Type	Description
deliveryPartnerId	ObjectId	Unique delivery ID
name	String	Delivery person name
phone	String	Contact number
status	String	Available / Busy

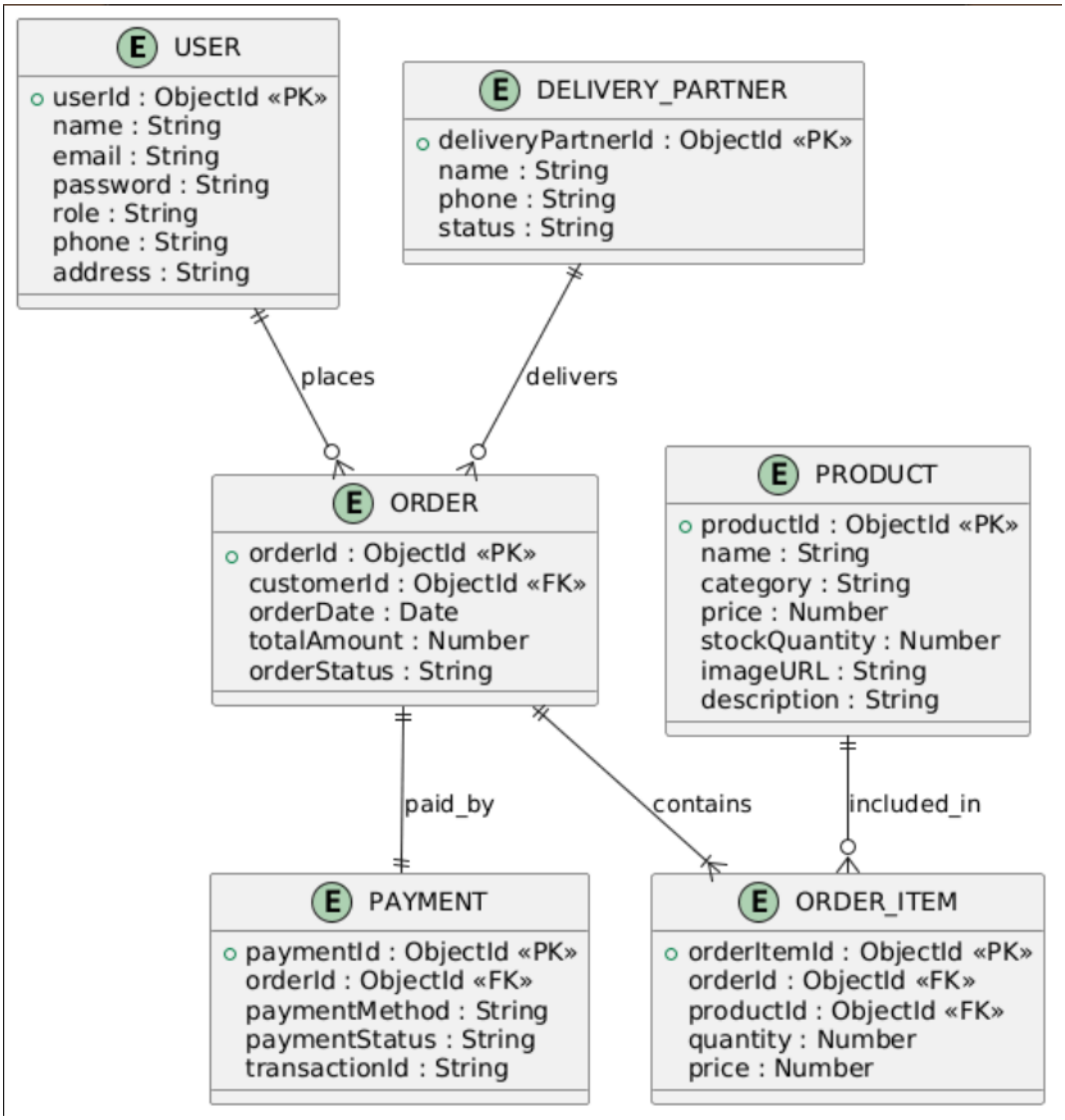


Fig 5.4 Database Relationship Diagram

5.3 INPUT/OUTPUT AND INTERFACE DESIGN

5.3.1 State Transition/UML Diagram

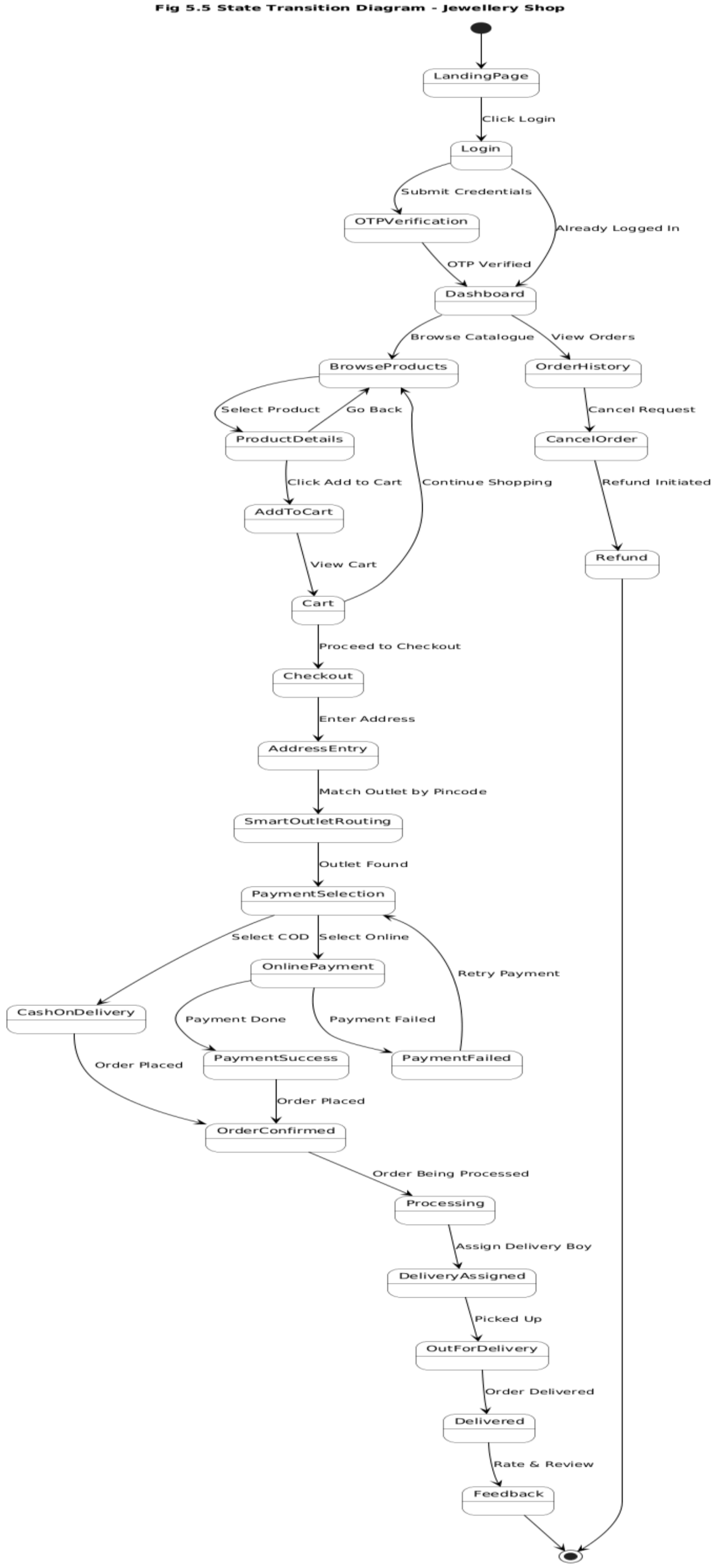


Fig 5.5 State Transition Diagram

6 IMPLEMENTATION PLANNING

6.1 IMPLEMENTATION ENVIRONMENT

The **Pariva Jewellery Shop system** is developed as a multi-user web-based e-commerce platform that allows customers to browse jewellery products, place orders, and track deliveries while enabling administrators and outlet managers to manage inventory and orders.

The system is implemented using modern web technologies that provide scalability, security, and responsive user interaction.

Table 6.1 Implementation Environment

Aspect	**Implementation Details**
System Type	Multi-user web-based application
Interface	Graphical User Interface (GUI)
Frontend	React.js with Vite
Backend	Node.js with Express.js (REST APIs)
Database	MongoDB Atlas
Hosting	Frontend deployed on Vercel/Netlify, Backend on Render/Railway
Security	JWT Authentication, Bcrypt Password Encryption
Payment Integration	Razorpay Payment Gateway
API Communication	Axios for REST API communication
Styling	CSS / Tailwind CSS

The system is accessible through modern web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. The responsive interface ensures compatibility across desktop, tablet, and mobile devices.

6.2 PROGRAM / MODULES SPECIFICATION

The Pariva Jewellery Shop system is divided into several modules that handle different functionalities of the platform.

Table 6.2 Core Modules

Module Name	Description	Technologies Used
User Authentication	Handles user registration, login, OTP verification, and role-based access control	Node.js, Express.js, JWT, Bcrypt
Product Management	Allows admin to add, update, delete, and manage jewellery products and categories	Node.js, MongoDB
Product Browsing	Enables customers to view jewellery products and filter by category, price, and material	React.js, Axios
Cart Management	Allows customers to add/remove products from cart and update quantities	React.js, Context API
Wishlist Module	Enables customers to save favourite products for future purchase	React.js
Checkout Module	Handles delivery address entry, coupon application, and order confirmation	React.js, Node.js
Order Management	Manages order creation, order history, order status updates	Node.js, MongoDB
Payment System	Processes secure online payments and handles Cash on Delivery	Razorpay API
Delivery Management	Assigns delivery personnel and tracks delivery status	Node.js, MongoDB
Notification System	Sends order confirmation and delivery notifications	Email API / Notification Service
Admin Dashboard	Allows administrators to manage users, orders, products, outlets, and reports	React.js, Node.js

Module Interaction Flow

The interaction between modules follows a structured workflow from product browsing to delivery completion.

1. User Authentication

The user registers or logs into the system using email and password. The system verifies the credentials and generates a JWT authentication token.

2. Product Browsing

Customers browse jewellery products categorized into rings, necklaces, bracelets, earrings, etc.

3. Cart Management

Customers add selected products to their cart and update quantities if necessary.

4. Wishlist Management

Customers can save favourite products in the wishlist for future purchases.

5. Checkout Process

Customers enter their delivery address, apply coupon codes, and review the order summary.

6. Order Creation

The system creates an order and stores it in the database.

7. Payment Processing

Customers select a payment method:

- Online payment using Razorpay
- Cash on Delivery (COD)

8. Order Processing

The outlet manager or admin verifies the order and prepares the products for delivery.

9. Delivery Assignment

The system assigns a Blue Dart to deliver the order.

10. Delivery Completion

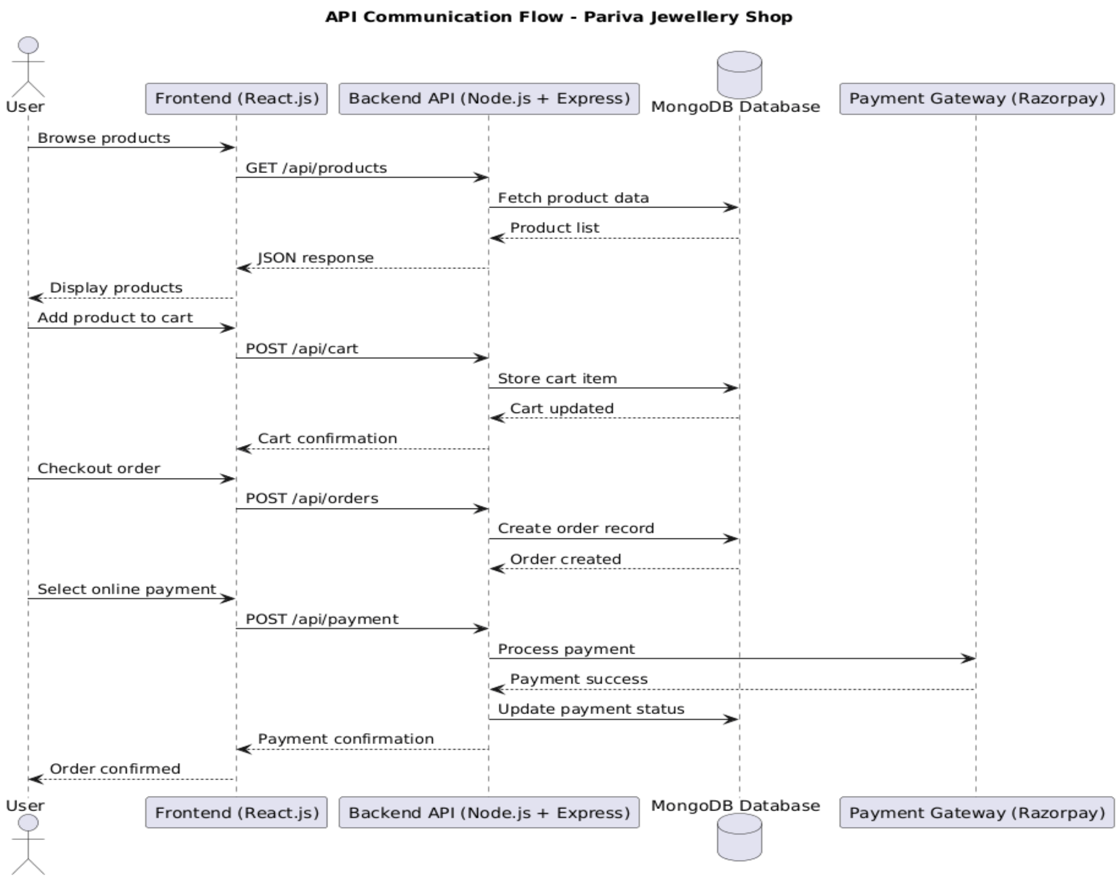
The delivery partner delivers the order and updates the delivery status to Delivered.

6.3 API DESIGN

Table 6.3 API Endpoints

Method	Endpoint	Description
POST	/api/auth/register	Register a new user
POST	/api/auth/login	User login
POST	/api/auth/verify-otp	Verify OTP authentication
GET	/api/products	Fetch all jewellery products
GET	/api/products/:id	Fetch product details
POST	/api/cart	Add product to cart
POST	/api/orders	Place a new order
GET	/api/orders	View order history
POST	/api/payment	Process payment
POST	/api/delivery	Assign delivery partner

Fig 6.1 api communication flow



7.TESTING

7.1 TESTING PLAN

Testing is an important phase in the development of the **Pariva Jewellery Shop system** to ensure that all modules work correctly and efficiently. The testing plan aims to verify that the application performs as expected and satisfies the system requirements.

The main objectives of the testing plan are:

- Identify and fix functional and logical errors in the system.
- Ensure secure authentication and safe handling of user data.
- Verify product browsing, cart management, and checkout functionalities.
- Validate payment processing through the payment gateway.
- Ensure proper communication between frontend and backend APIs.
- Verify order processing and delivery management.
- Ensure the system works across different browsers and devices.

Testing was conducted at different levels to ensure that each module of the system functions correctly before integrating it with other modules.

7.2 TESTING STRATEGY

The testing strategy adopted for the Pariva Jewellery Shop system includes multiple testing approaches to ensure the reliability and performance of the system.

Unit Testing

Each individual component or module such as authentication, product management, cart management, and order processing was tested separately to verify that it works correctly.

Integration Testing

Integration testing was performed to ensure smooth interaction between modules such as:

Product Browsing → Cart → Checkout → Payment → Order Confirmation

This ensures that data flows correctly between different parts of the system.

System Testing

The complete system was tested as a whole to verify that all features work together without errors. End-to-end user scenarios were tested including browsing products, placing orders, and completing payments.

User Acceptance Testing (UAT)

Realistic user scenarios were tested to ensure that the system meets the expectations of customers and administrators. Feedback was collected to improve usability and functionality.

7.3 TESTING METHODS

Table 7.1 Testing Methods

Method	Description
Manual Testing	User interface and functional features were tested manually by simulating different user actions such as login, product browsing, and order placement.
Responsive Testing	The system interface was tested across desktop, tablet, and mobile devices to ensure responsive design.
API Testing	Backend APIs were tested using Postman before integration with the frontend.
Security Testing	Authentication mechanisms including JWT tokens, password encryption, and secure API communication were tested.
Performance Testing	System response time for product search, order placement, and payment processing was evaluated.

7.4 TEST CASES

7.4.1 Purpose

The purpose of test cases is to verify that each major function of the system works correctly and does not affect other functionalities.

Test cases ensure that the application behaves as expected under different input conditions.

7.4.2 Required Input

The following inputs were used during testing:

- Valid and invalid login credentials
- Product search keywords
- Cart operations and quantity updates
- Checkout details and delivery address
- Payment gateway information

- Order processing actions by admin

7.4.3 Expected Results

Table 7.2 Test Cases and Expected Results

Test Case	**Input**	**Expected Output**	**Actual Output**	**Pass/Fail**
User Login	Valid email and password	User logged into dashboard	Dashboard displayed	✓
Product Search	Product name or category	Relevant products displayed	Products shown correctly	✓
Add to Cart	Select product and quantity	Product added to cart	Product added successfully	✓
Checkout Process	Address and order confirmation	Order created and stored in database	Order created successfully	✓
Online Payment	Valid payment details	Payment successful and order confirmed	Payment completed	✓
Cash on Delivery	Select COD option	Order placed without online payment	Order confirmed	✓
Delivery Assignment	Assign delivery partner	Delivery status updated	Delivery assigned successfully	✓
Order Tracking	Check order status	Updated delivery status displayed	Status updated correctly	✓

8 USER MANUAL

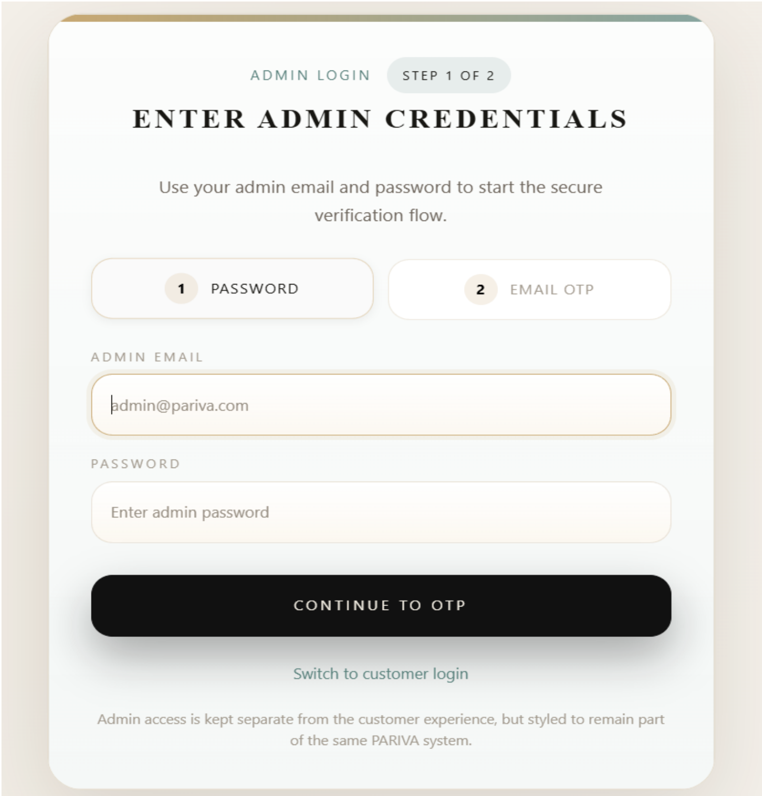


Fig 8.1 Admin Login Page

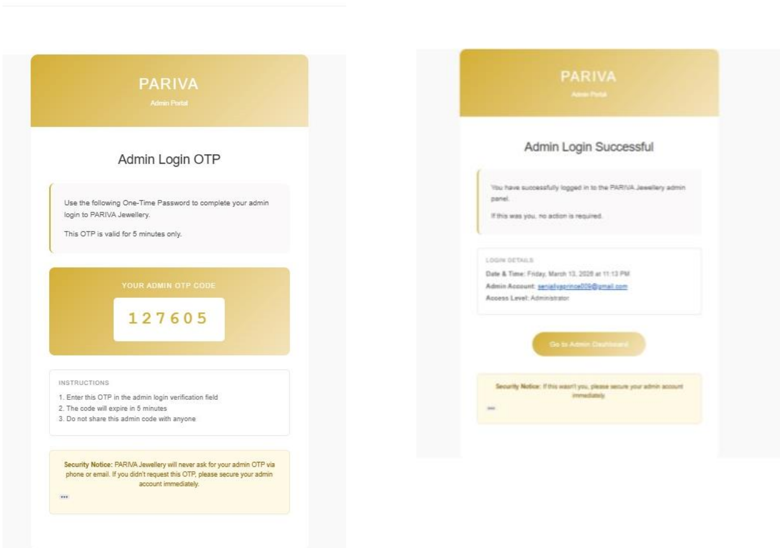


Fig 8.2 Authentication via OTP

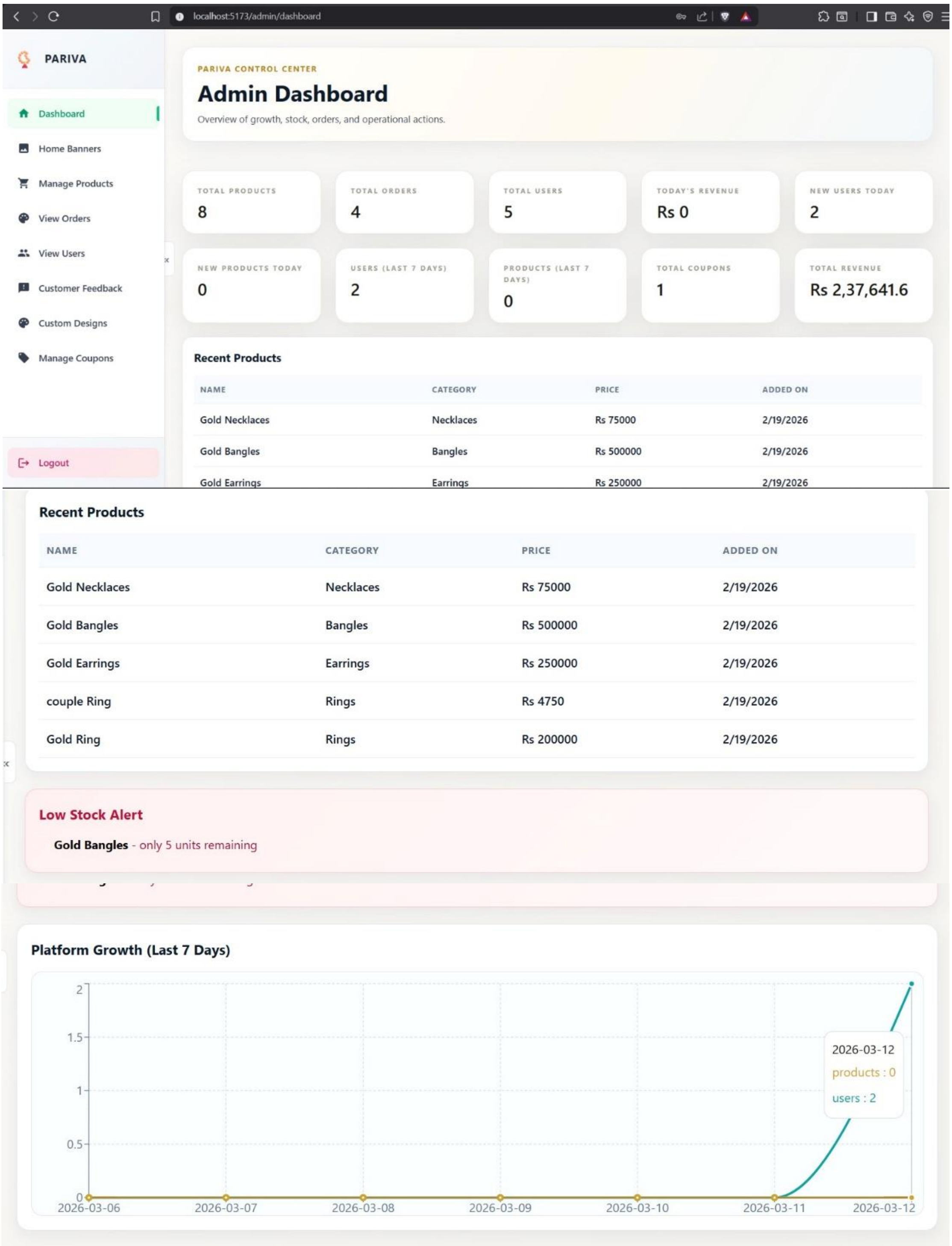


Fig 8.3 Admin Dashboard

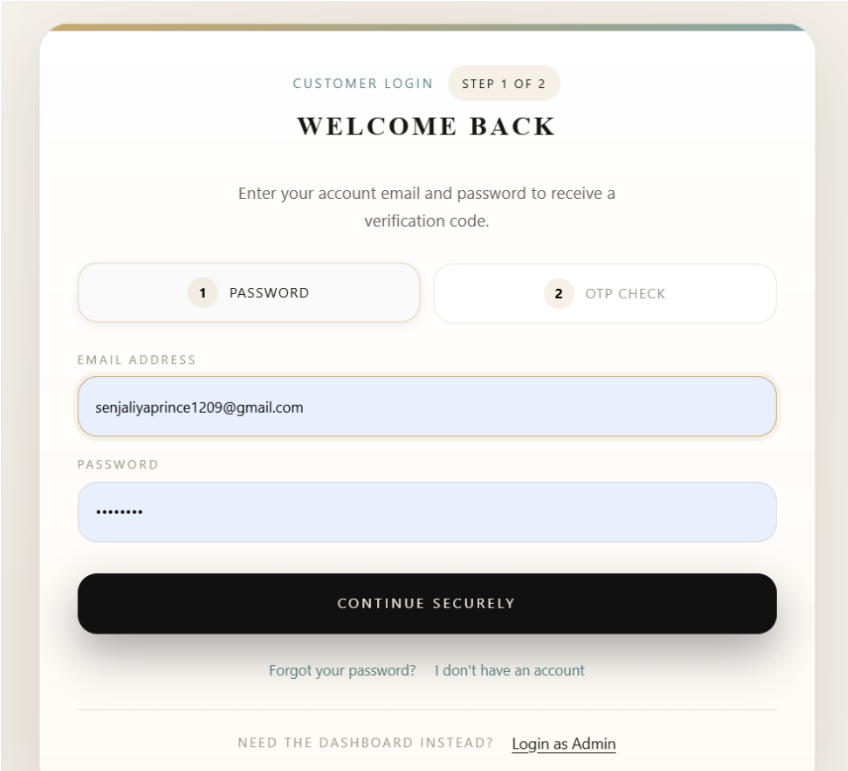


Fig 8.4 User Login

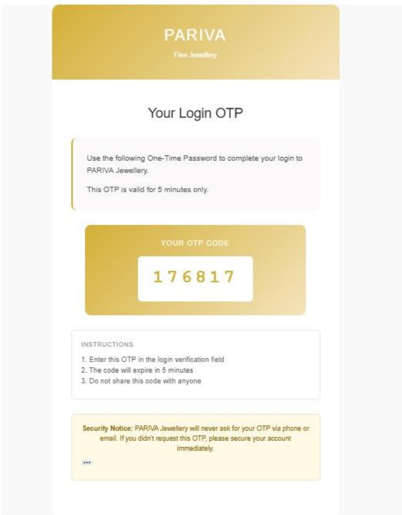
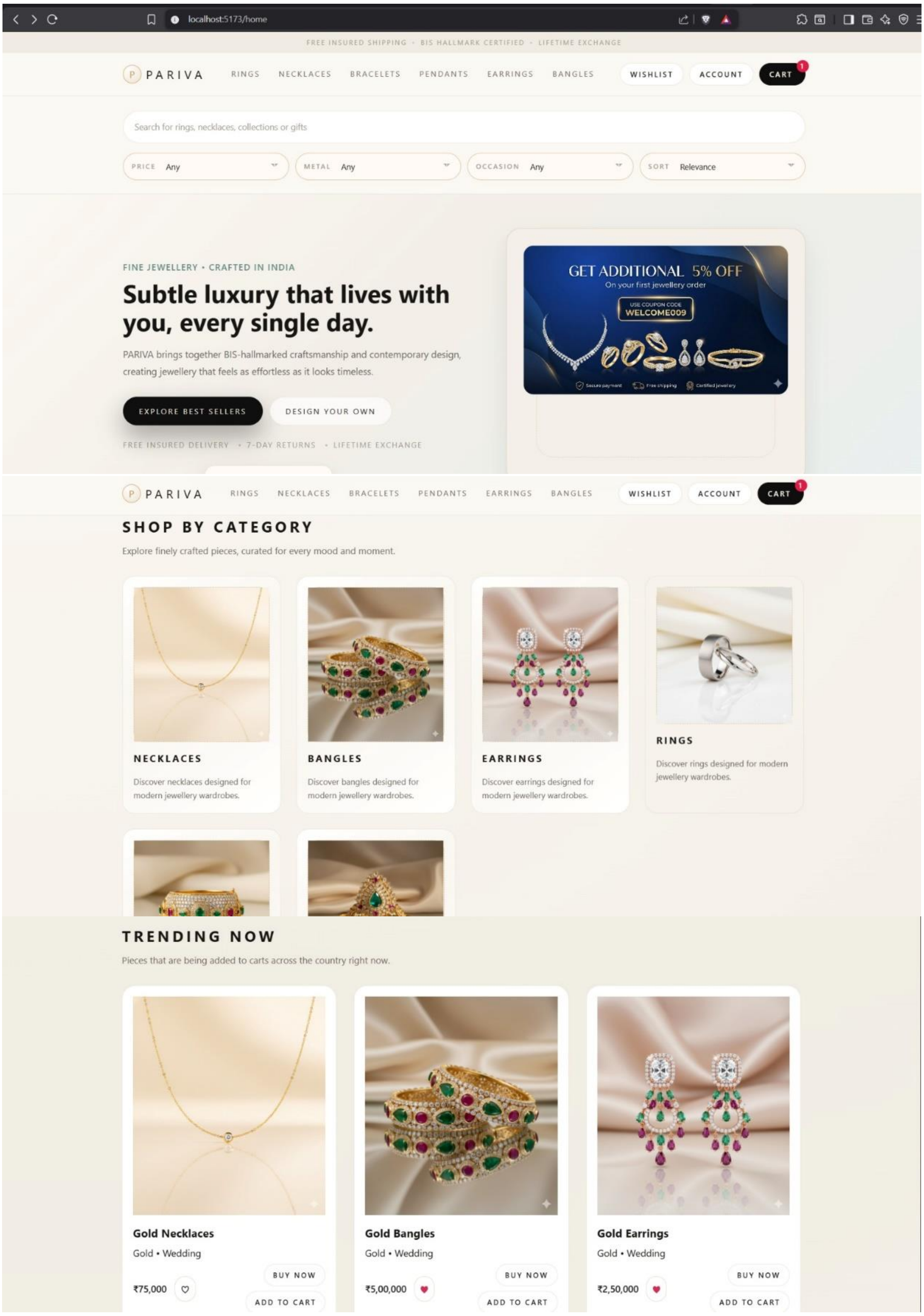


Fig 8.5 Authentication via OTP


localhost:5173/home
FREE INSURED SHIPPING • BIS HALLMARK CERTIFIED • LIFETIME EXCHANGE
PARIVA
RINGS
NECKLACES
BRACELETS
PENDANTS
EARRINGS
BANGLES
WISHLIST
ACCOUNT
CART
Search for rings, necklaces, collections or gifts
PRICE Any
METAL Any
OCCASION Any
SORT Relevance
FINE JEWELLERY • CRAFTED IN INDIA
Subtle luxury that lives with you, every single day.
PARIVA brings together BIS-hallmarked craftsmanship and contemporary design, creating jewellery that feels as effortless as it looks timeless.
EXPLORE BEST SELLERS
DESIGN YOUR OWN
FREE INSURED DELIVERY • 7-DAY RETURNS • LIFETIME EXCHANGE
GET ADDITIONAL 5% OFF
On your first jewellery order
USE COUPON CODE
WELCOME009
SHOP BY CATEGORY
Explore finely crafted pieces, curated for every mood and moment.
NECKLACES
Discover necklaces designed for modern jewellery wardrobes.
BANGLES
Discover bangles designed for modern jewellery wardrobes.
EARRINGS
Discover earrings designed for modern jewellery wardrobes.
RINGS
Discover rings designed for modern jewellery wardrobes.
TRENDING NOW
Pieces that are being added to carts across the country right now.
Gold Necklaces
Gold • Wedding
₹75,000
BUY NOW
ADD TO CART
Gold Bangles
Gold • Wedding
₹5,00,000
BUY NOW
ADD TO CART
Gold Earrings
Gold • Wedding
₹2,50,000
BUY NOW
ADD TO CART

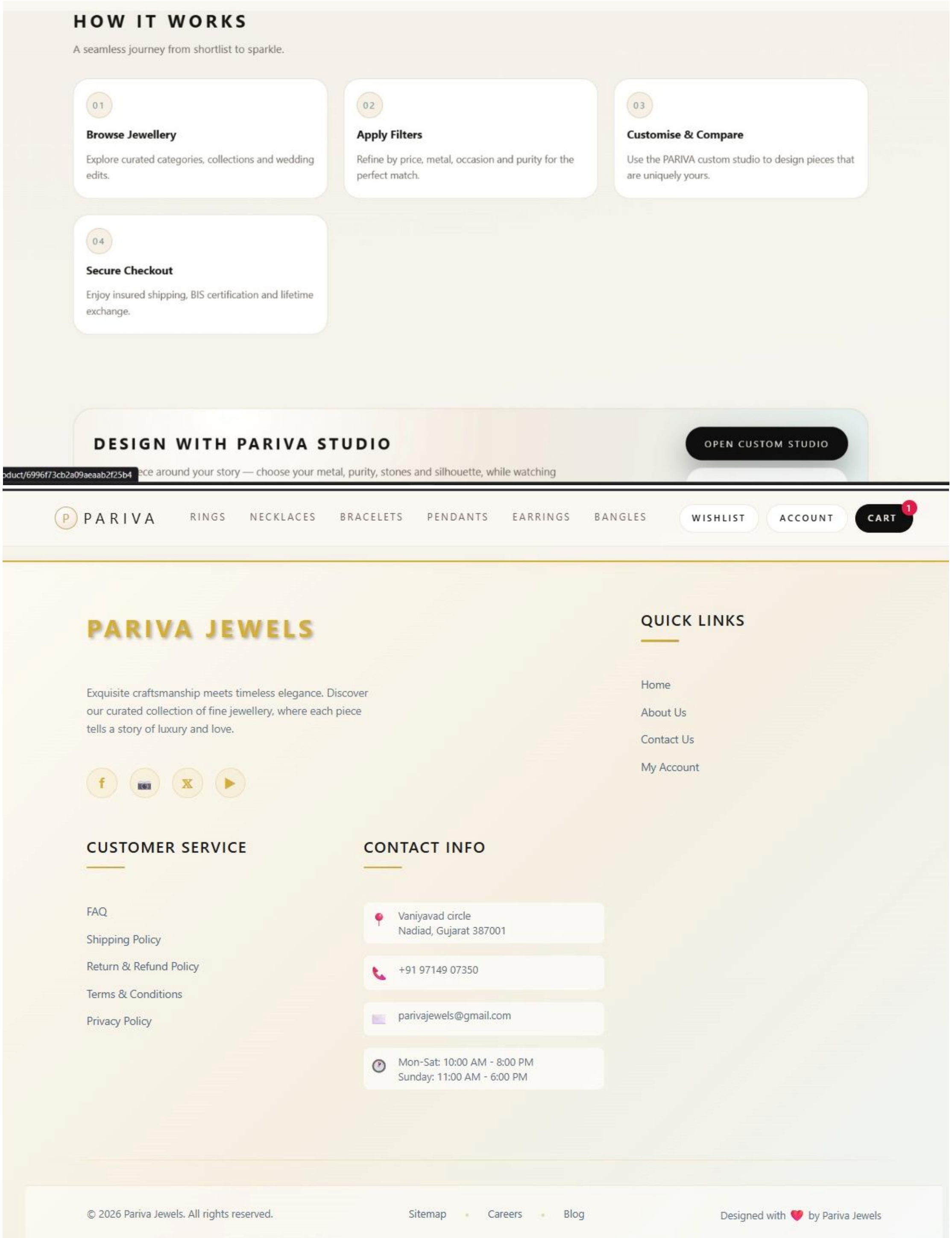
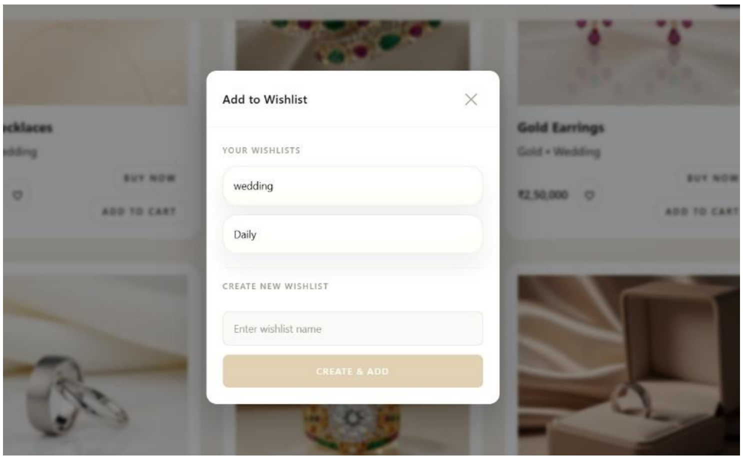


Fig 8.6 User Dashboard



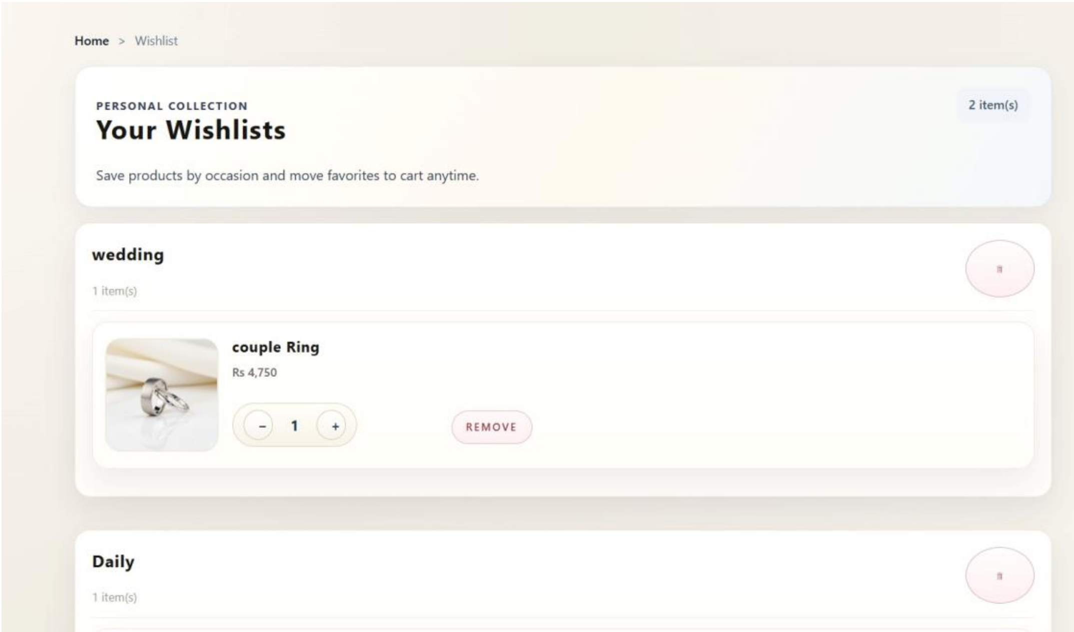


Fig 8.7 Wishlist

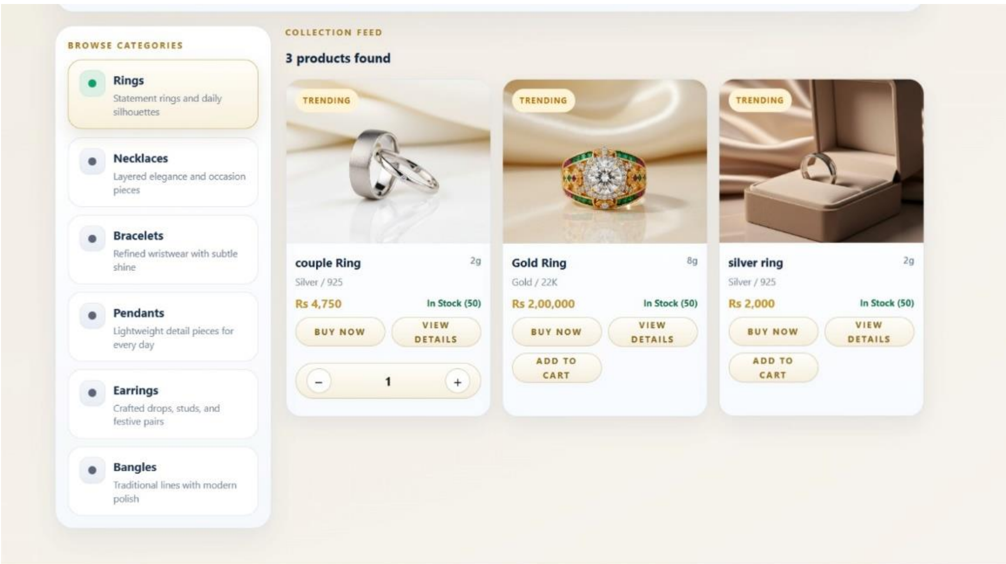


Fig.8.8 Product List

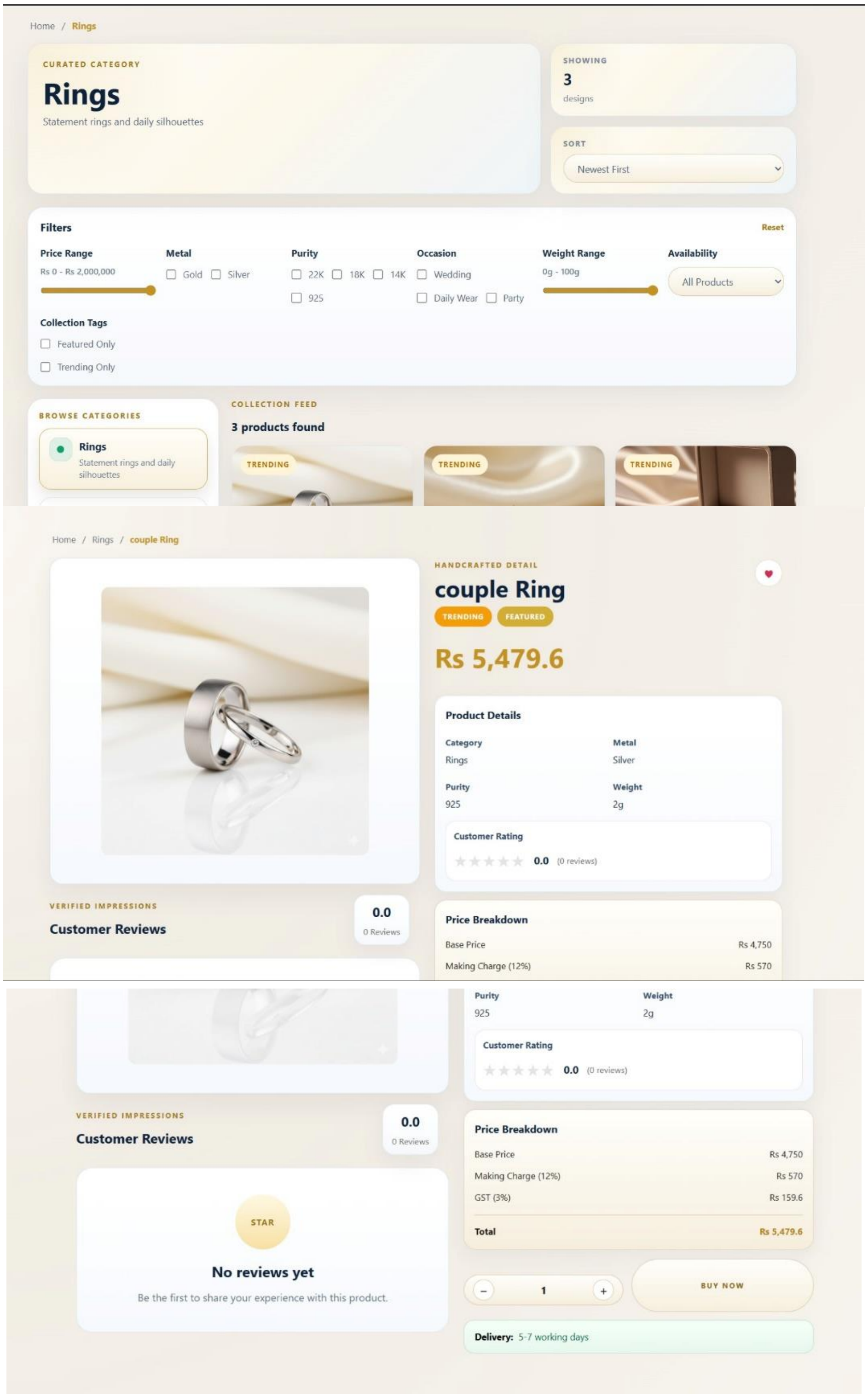


Fig 8.9 Product Details

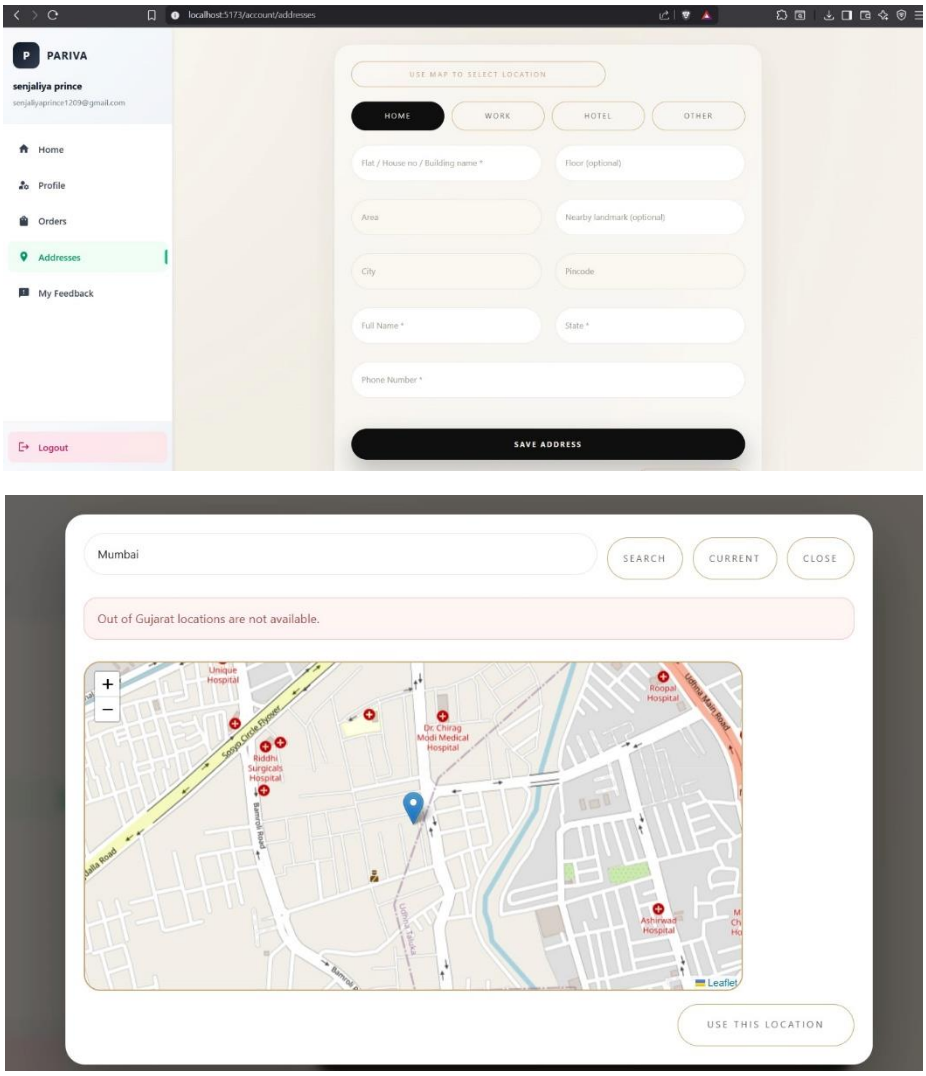


Fig 8.10 Address details

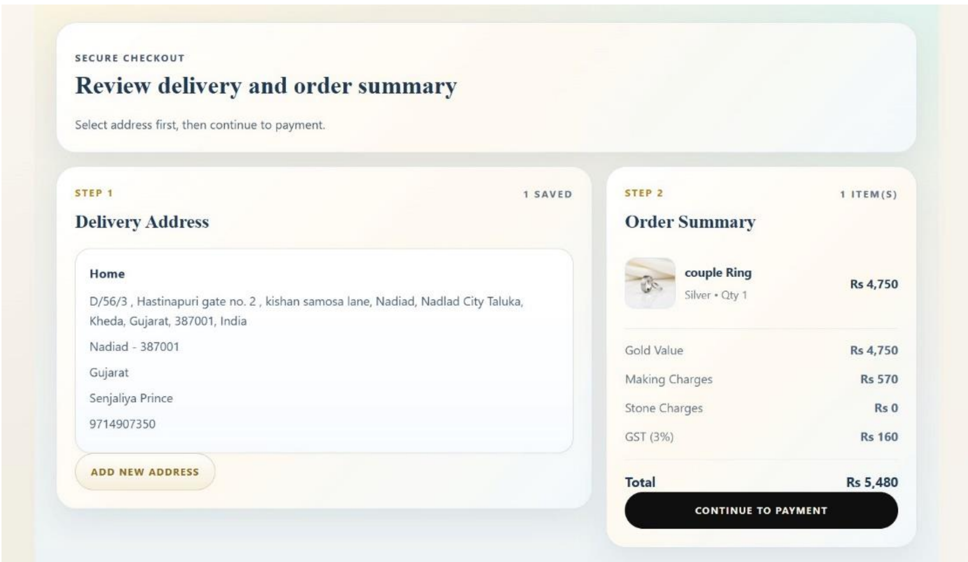
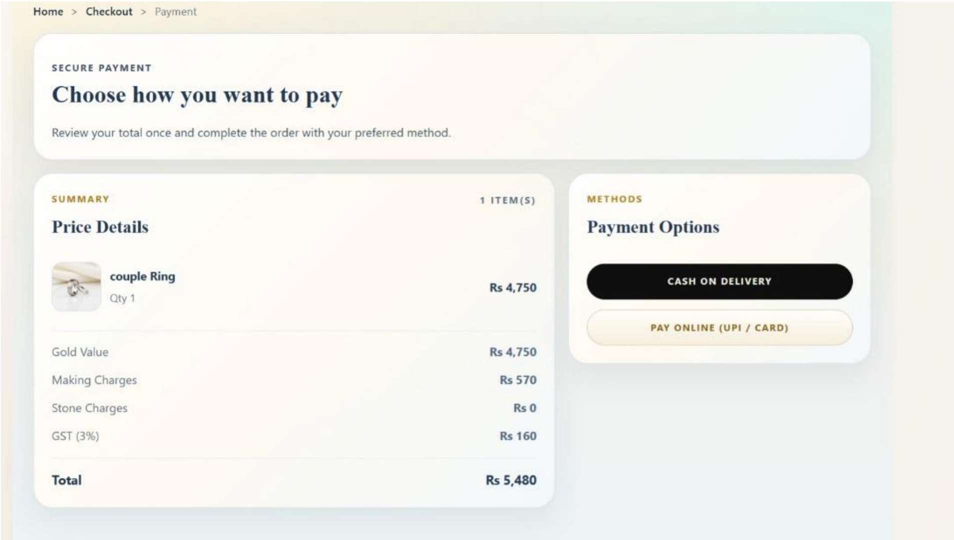


Fig 8.11 Checkout



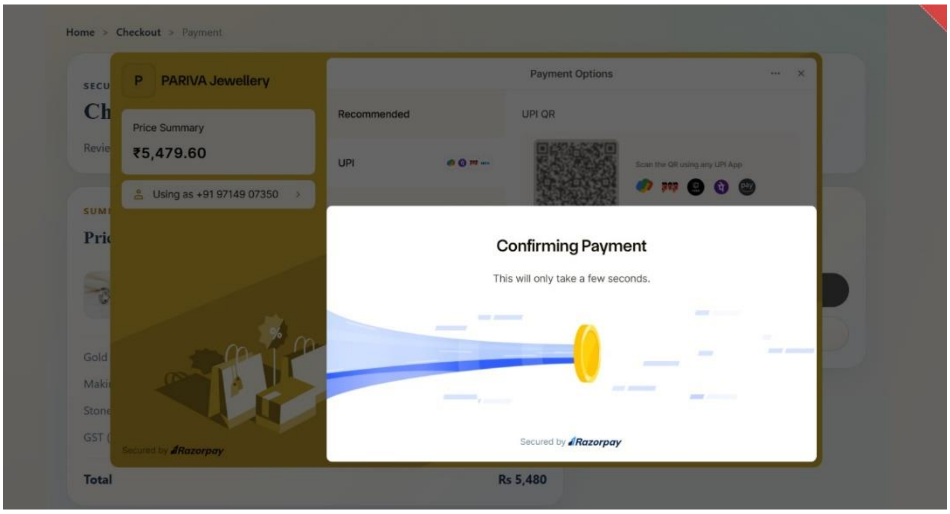


Fig 8.12 Payment

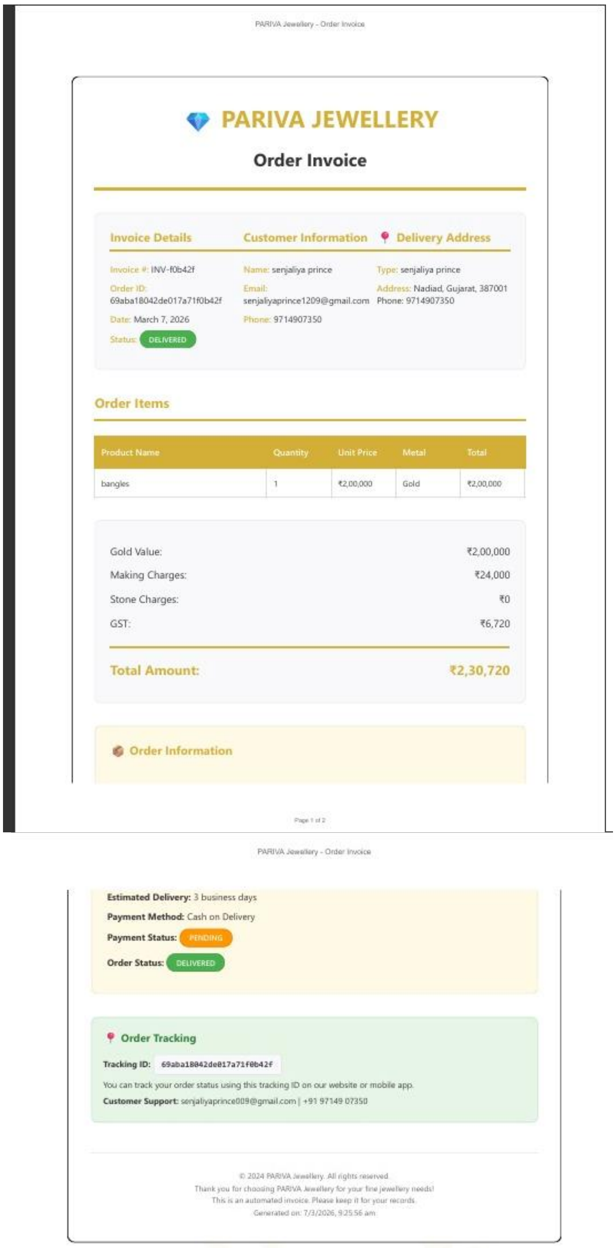


Fig 8.13 Invoice

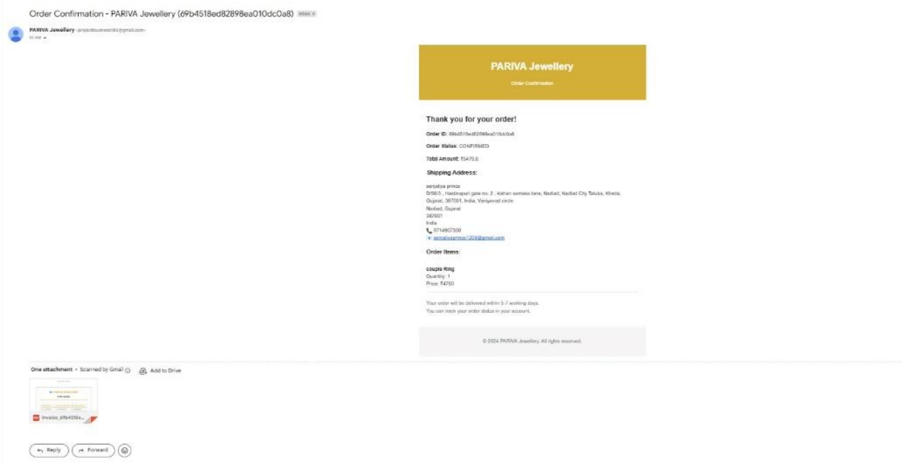


Fig 8.14 Order Confirmation

9. LIMITATIONS AND FUTURE ENHANCEMENTS

9.1 CURRENT LIMITATIONS

Although the **Pariva Jewellery Shop system** provides a functional and efficient online platform for browsing jewellery products and placing orders, some limitations still exist in the current version of the system.

1. Limited Payment Options

The system currently supports online payments through Razorpay and Cash on Delivery (COD). However, integration with additional regional payment services could improve accessibility and user convenience.

2. Basic Inventory Management

The system provides basic product stock management for outlets and administrators. Advanced inventory features such as automatic stock prediction or demand forecasting are not currently implemented.

3. No Real-Time Delivery Tracking

Although customers can view order status updates such as order placed, processing, and delivered, the system does not currently support GPS-based real-time delivery tracking.

4. Limited Personalization

The system currently displays products to users without personalized recommendations based on browsing history or previous purchases.

5. Internet Dependency

Since the system is a web and app based platform , users must have an active internet connection to browse products and place orders.

9.2 FUTURE ENHANCEMENTS

The Pariva Jewellery Shop platform can be enhanced further by implementing additional advanced features in future versions.

1. AI-Based Product Recommendation

Machine learning algorithms can be used to analyze customer browsing behavior and purchase history to recommend relevant jewellery products.

2. Real-Time Delivery Tracking

Integration of GPS-based delivery tracking can allow customers to monitor the real-time location of delivery personnel.

3. Advanced Inventory Management

Predictive analytics can be used to forecast product demand and automatically manage stock levels for outlets.

4. Multi-Language Support

Adding support for multiple languages such as Gujarati, Hindi, and English would make the platform more accessible to users from different regions.

5. Customer Review and Rating System

Future versions can include a review and rating system where customers can share feedback about products and services.

6. Loyalty and Reward Programs

Introducing loyalty programs, reward points, and discount offers can improve customer retention and increase platform engagement.

10. CONCLUSION AND DISCUSSION

10.1 Project Achievements

The **Pariva Jewellery Shop platform** successfully implements the core functionalities required for a modern online jewellery shopping system. The project demonstrates how digital technologies can be used to simplify the process of browsing jewellery products, placing orders, and managing deliveries.

Successful Implementation of Core Features

The system successfully implemented several important features required for an online jewellery marketplace.

- Customers can easily browse jewellery products categorized as rings, necklaces, bracelets, earrings, and other jewellery items.
- Secure user authentication using JWT and password encryption ensures safe user login and account management.
- The cart and checkout system allows customers to select products and place orders efficiently.
- Payment integration through Razorpay enables secure online payments along with a Cash on Delivery option.
- Order management allows administrators and outlet managers to process and track orders.
- Delivery management enables assignment of delivery personnel and updating delivery status.

Robust System Architecture

The Pariva Jewellery Shop platform follows a **three-tier architecture**, ensuring scalability and efficient system performance.

- The frontend is developed using React.js to provide a dynamic and responsive user interface.
- The backend is implemented using Node.js and Express.js, which handle business logic and REST API communication.
- The database uses MongoDB to store user information, product data, and order records.
- Secure authentication mechanisms such as JWT tokens and encrypted passwords help protect user information.

Scalable System Design

The system is designed to support future expansion and handle increasing numbers of users.

- The database structure allows efficient storage and retrieval of product and order information.
- Modular architecture allows new features to be added without affecting existing functionalities.
- Cloud-based deployment ensures that the system can scale according to user demand.

User-Friendly Interface

The user interface of the system is designed to be simple and easy to use.

- Customers can browse products and place orders with minimal effort.
- Clear product displays and filtering options improve the product discovery process.
- The responsive design ensures compatibility across desktops, tablets, and mobile devices.

10.2 Lessons Learned

Technical Insights

During the development of the Pariva Jewellery Shop system, several technical insights were gained.

- Designing a well-structured database is essential for managing product inventory and orders efficiently.
- Implementing secure authentication mechanisms is important for protecting user accounts and transactions.
- Developing scalable APIs helps maintain system performance and simplifies integration with frontend applications.
- Using modern frameworks such as React improves the user experience and simplifies interface development.

Development Challenges

Several challenges were encountered during the development process.

- Integrating multiple modules such as authentication, product management, cart, and payment systems required careful planning.
- Ensuring smooth communication between frontend and backend APIs required proper error handling.

- Managing product inventory and order data in the database required efficient schema design.
- Testing the application across different devices and browsers required extensive validation.

Improvement Areas

Although the system functions successfully, some improvements can be considered in future development.

- System documentation can be expanded to assist future developers.
- Automated testing can be introduced to improve system reliability.
- Performance optimization can further enhance user experience.
- Customer feedback can be used to continuously improve the platform.

10.3 Future Outlook

Expansion Opportunities

The Pariva Jewellery Shop platform has strong potential for future development and expansion.

- The platform can expand its jewellery catalog with more product categories and designs.
- Integration with additional payment gateways can improve user convenience.
- Development of a mobile application can increase accessibility and reach more customers.
- Integration with analytics tools can help analyze customer behavior and improve sales strategies.

Business Potential

The Pariva Jewellery Shop platform demonstrates how digital technologies can transform traditional jewellery retail businesses.

- Customers benefit from easy access to jewellery products through an online platform.
- Jewellery outlets can manage product inventory and orders efficiently.
- The platform can grow into a complete digital jewellery marketplace.

By combining modern web technologies with efficient system design, the **Pariva Jewellery Shop system** provides a reliable and scalable solution for online jewellery retail management.

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